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Chang

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(54) **CLAMP-ON TYPE PIPELINE MEASURING DEVICES AND FIXING STRUCTURES OF THE SAME**

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G01B 5/00 (2006.01)

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USPC 33/412, 529, 613, 645
See application file for complete search history.

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Primary Examiner — George B Bennett

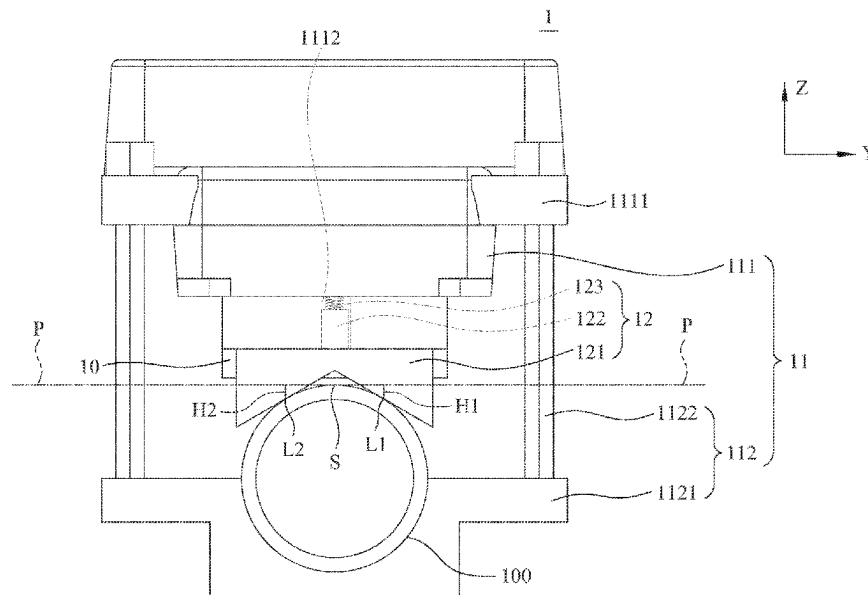
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(57)

ABSTRACT

Provided is a fixing structure of a clamp-on type pipeline measuring device, which includes a clamp and an aligning member, the clamp is detachably fixed on an outer surface of a pipeline along an axis on the outer surface of the pipeline, the aligning member is disposed in the clamp, wherein a plane tangent to the outer surface of the pipeline on the axis is defined as an installation datum plane. During detaching and fixing the clamp, the aligning member moves along a normal direction of the installation datum plane; after the clamp is fixed to the outer surface of the pipeline, the aligning member contacts at least two line segments on the outer surface of the pipeline, and distances between the two line segments and the installation datum plane are the same. The clamp-on type pipeline measuring device including the fixing structure is further provided.

10 Claims, 26 Drawing Sheets



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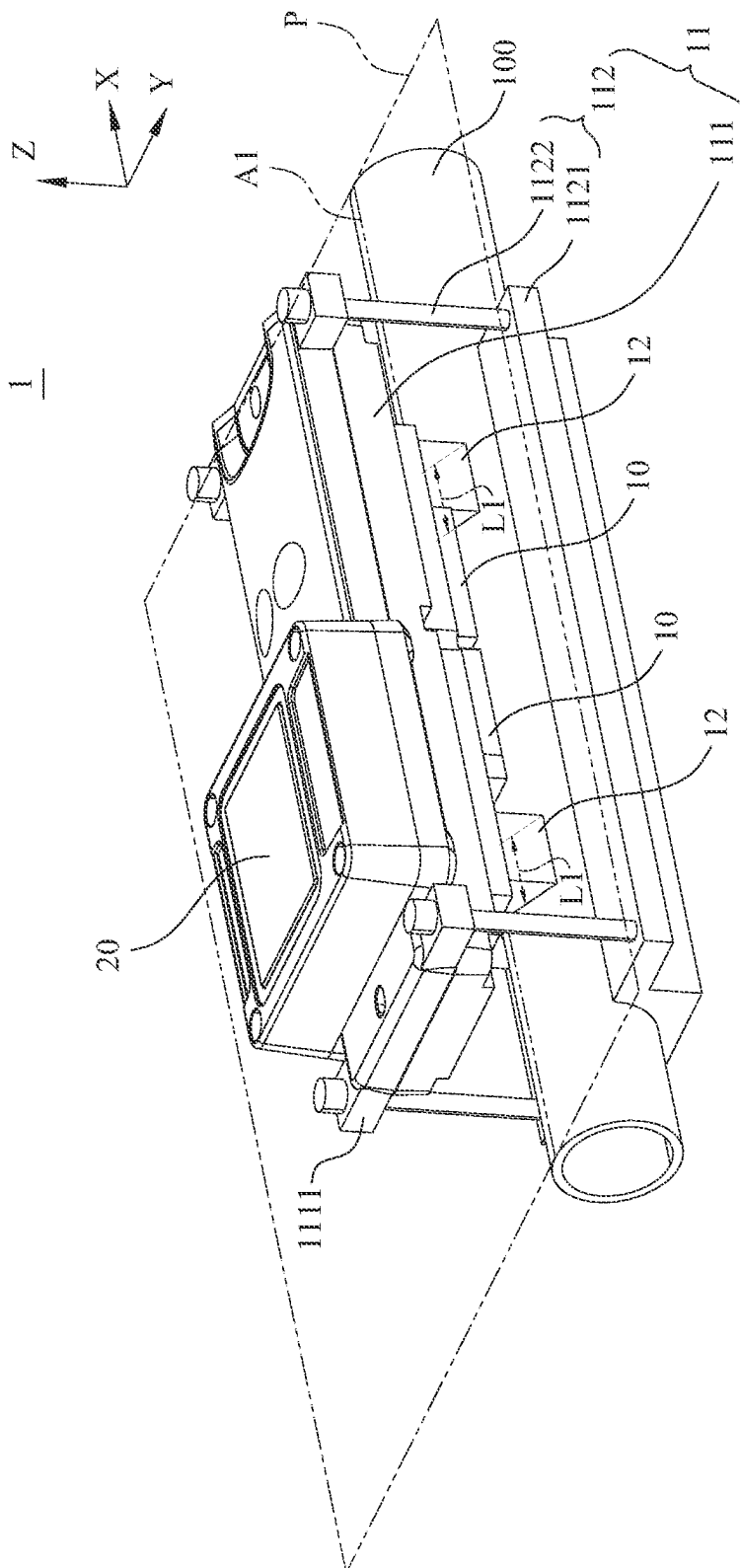


FIG. 1A

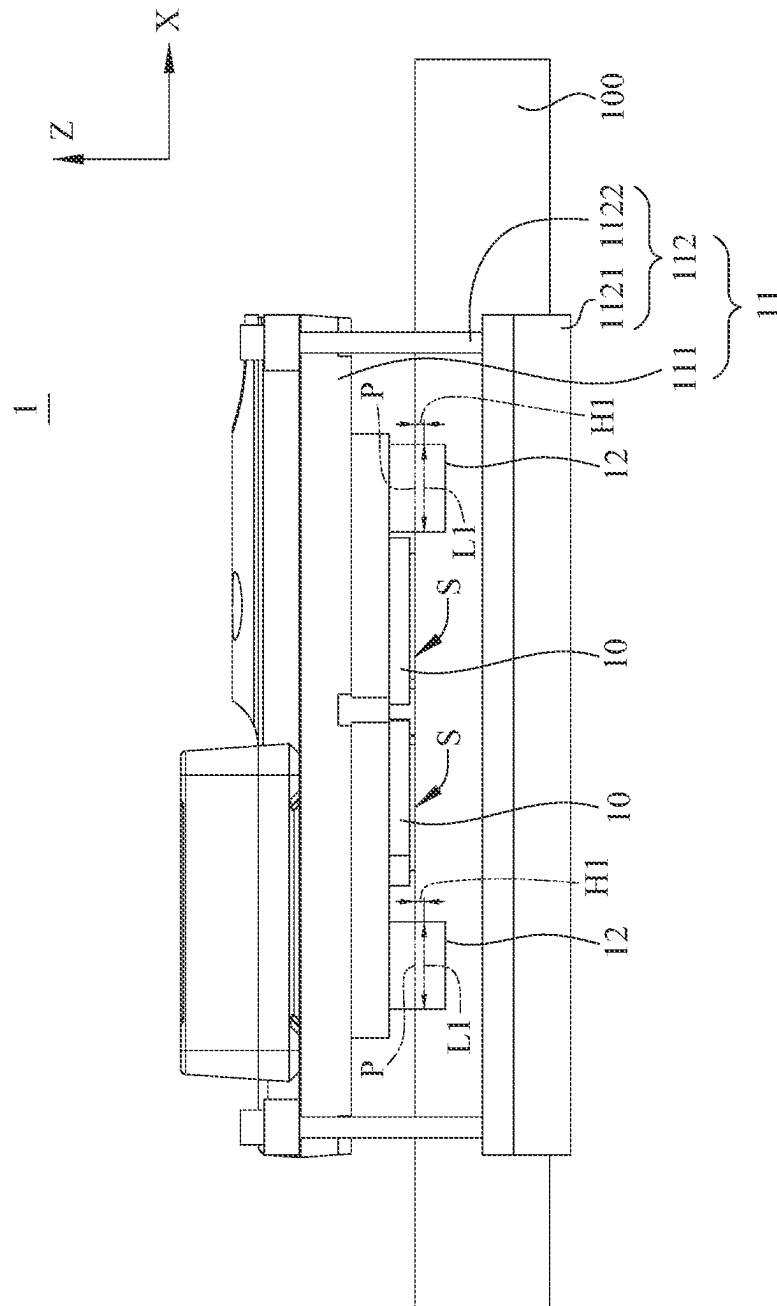


FIG. 13

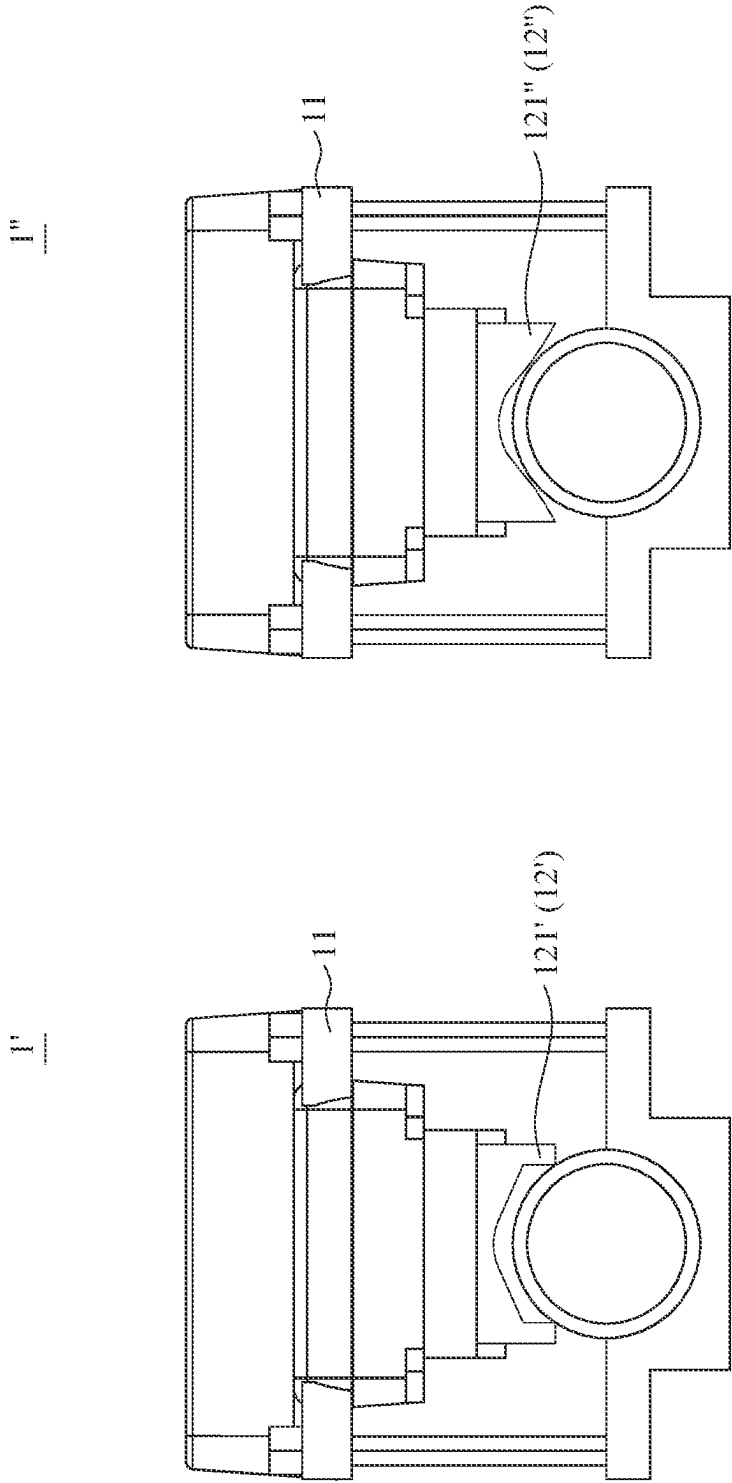


FIG. 1D

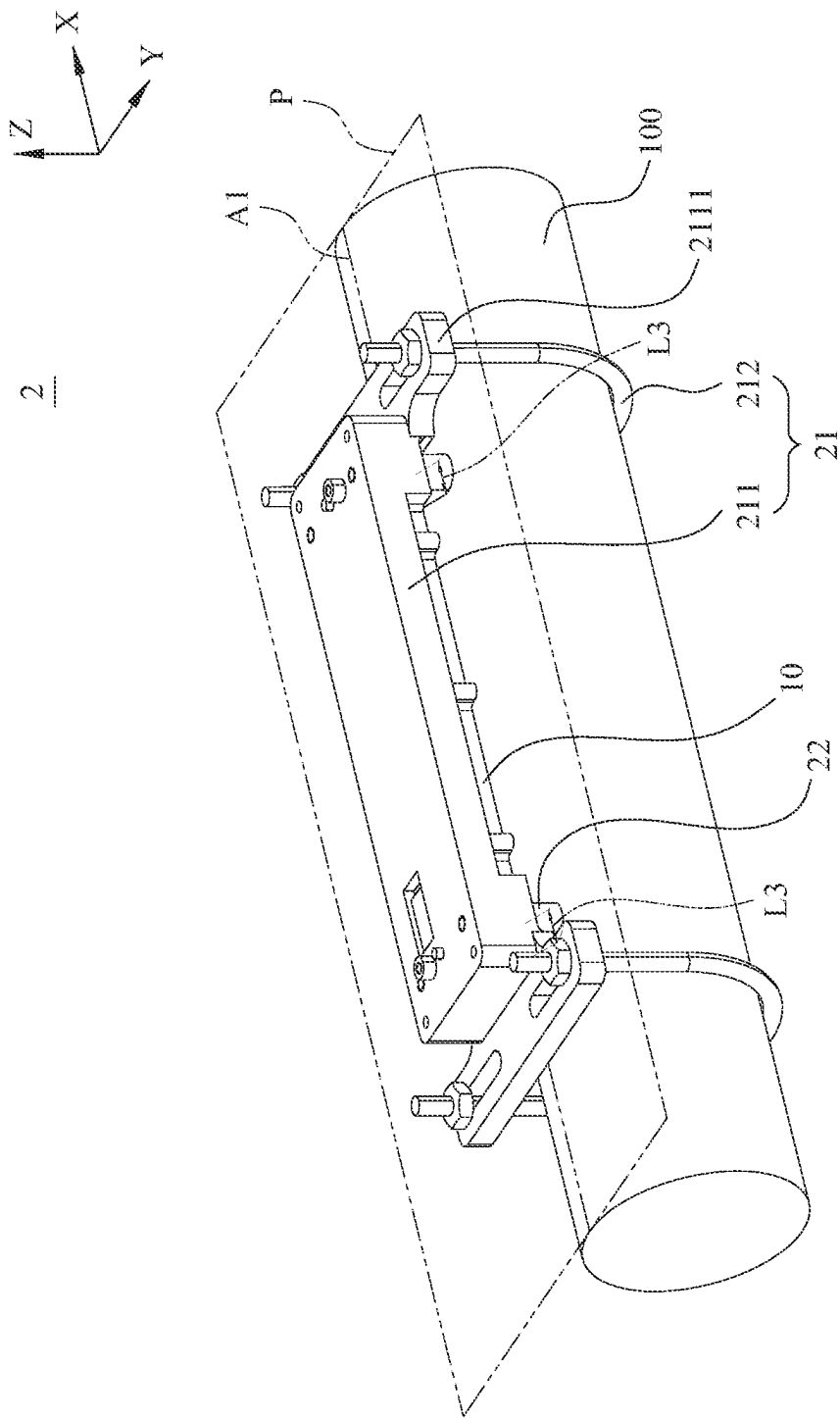


FIG. 2A

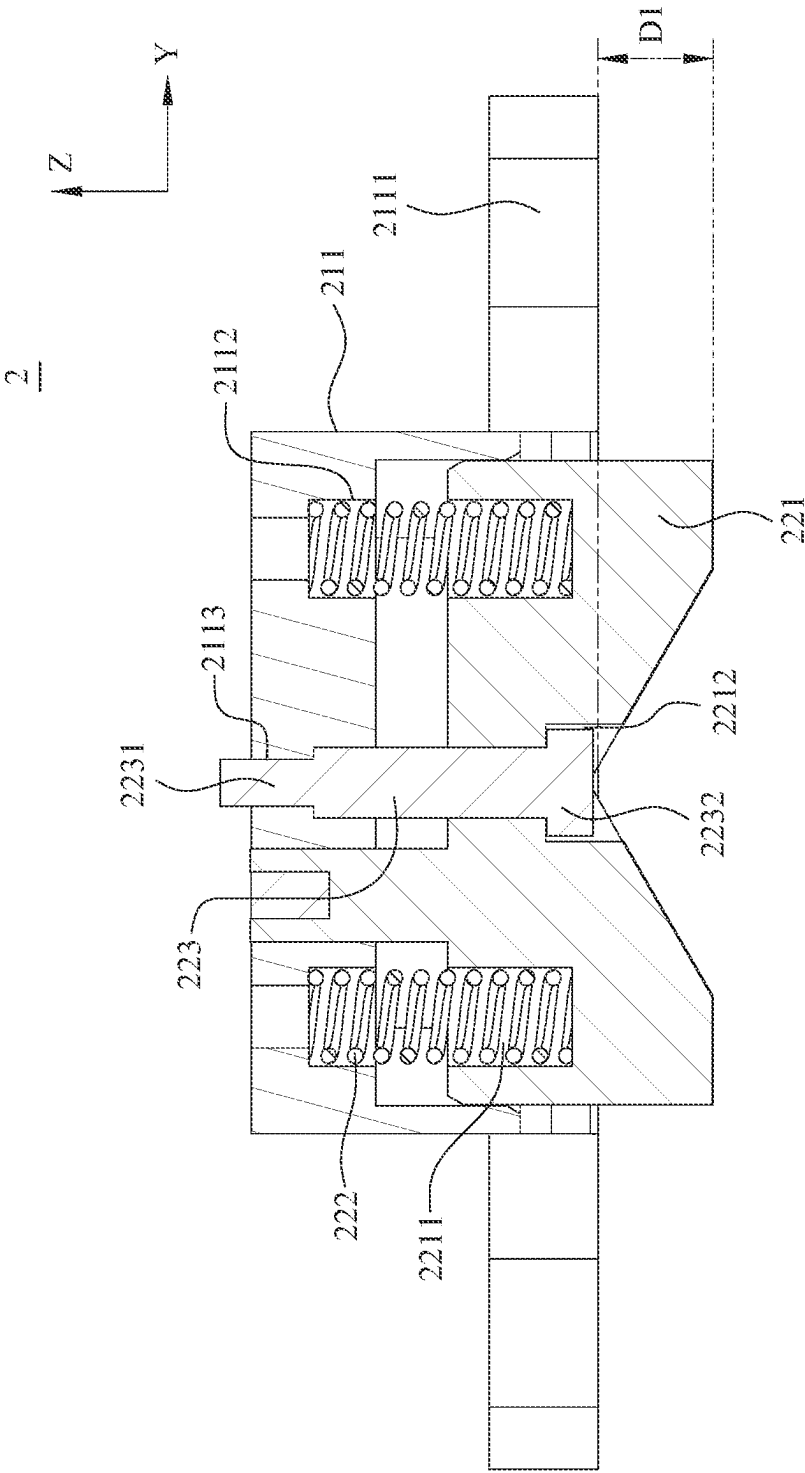


FIG. 2B

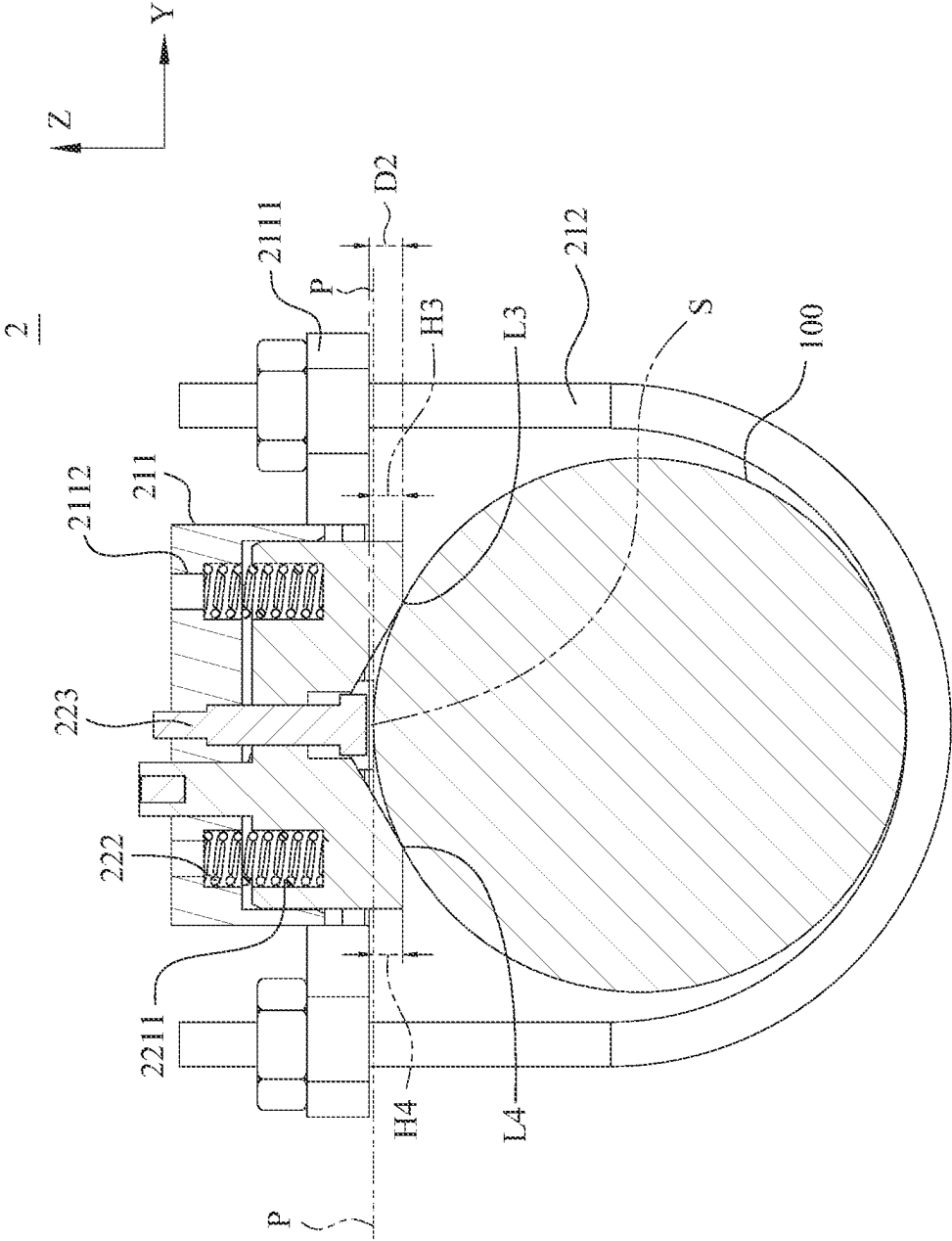


FIG. 2C

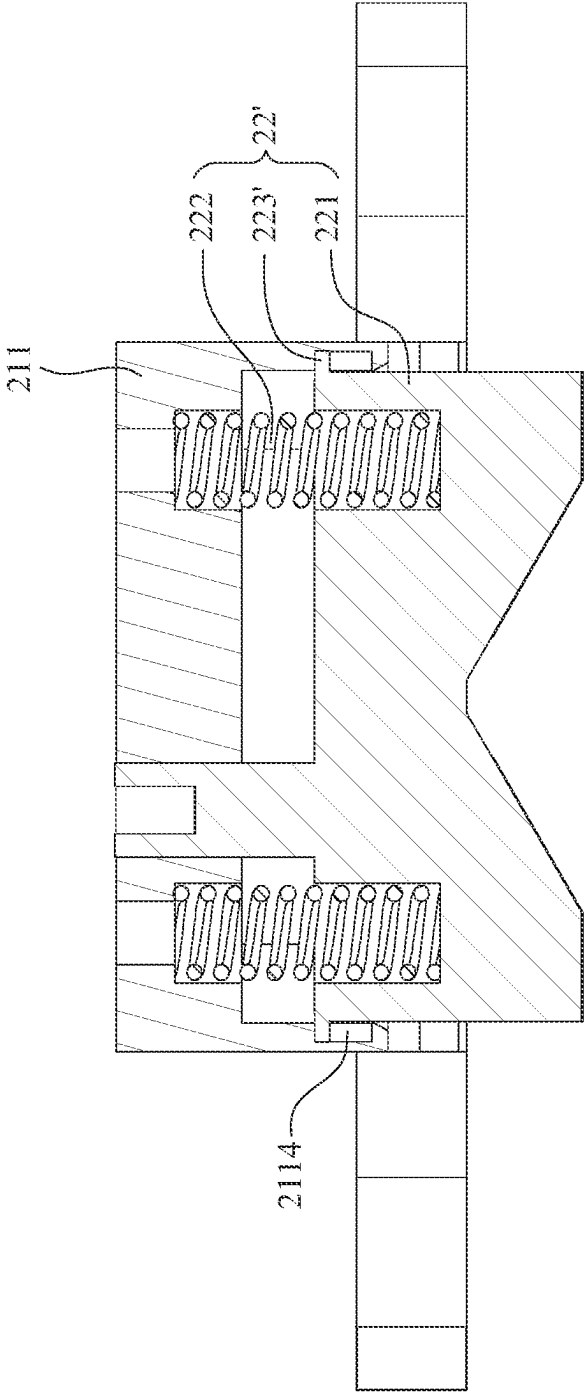


FIG. 2D

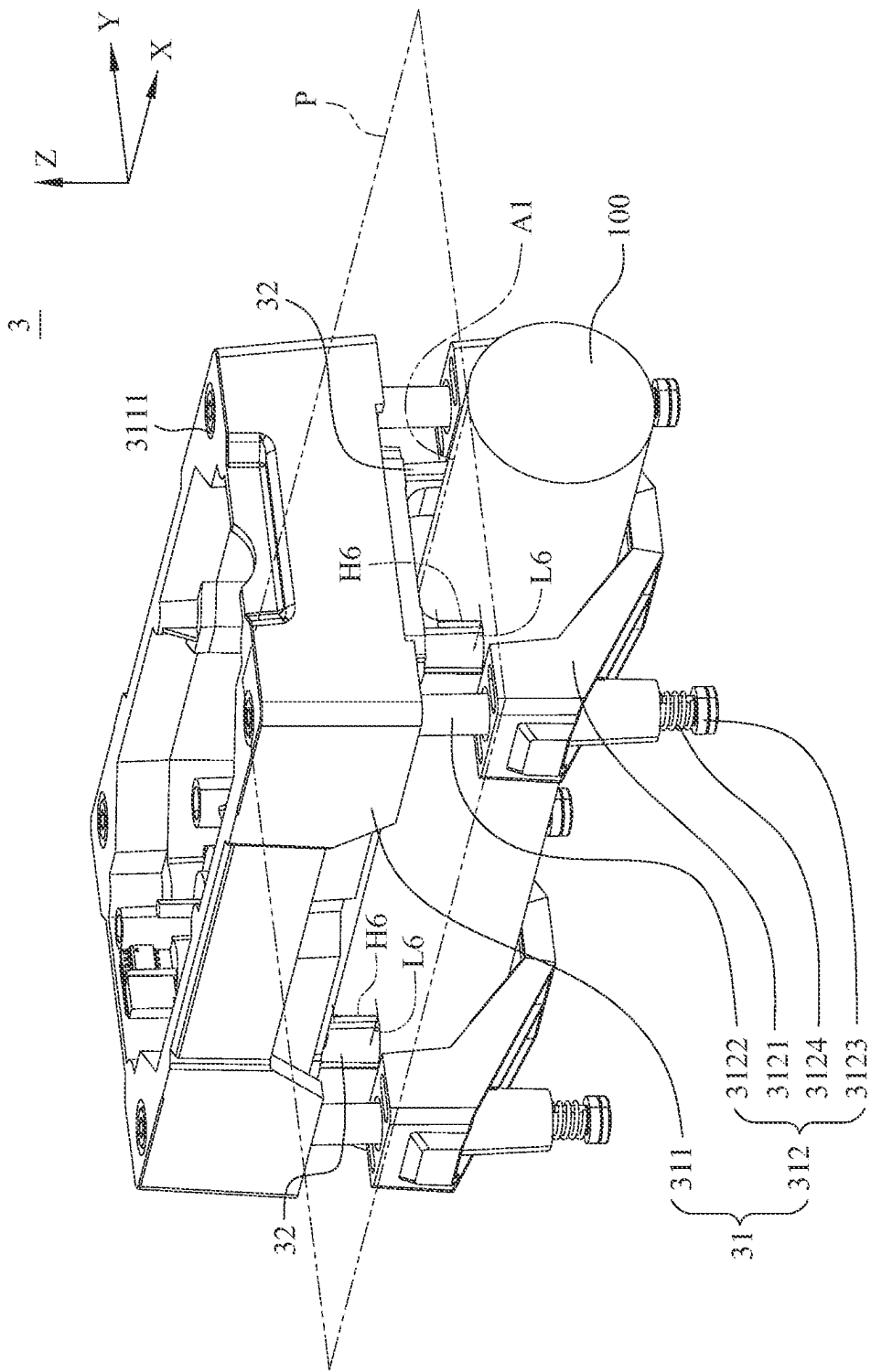


FIG. 3A

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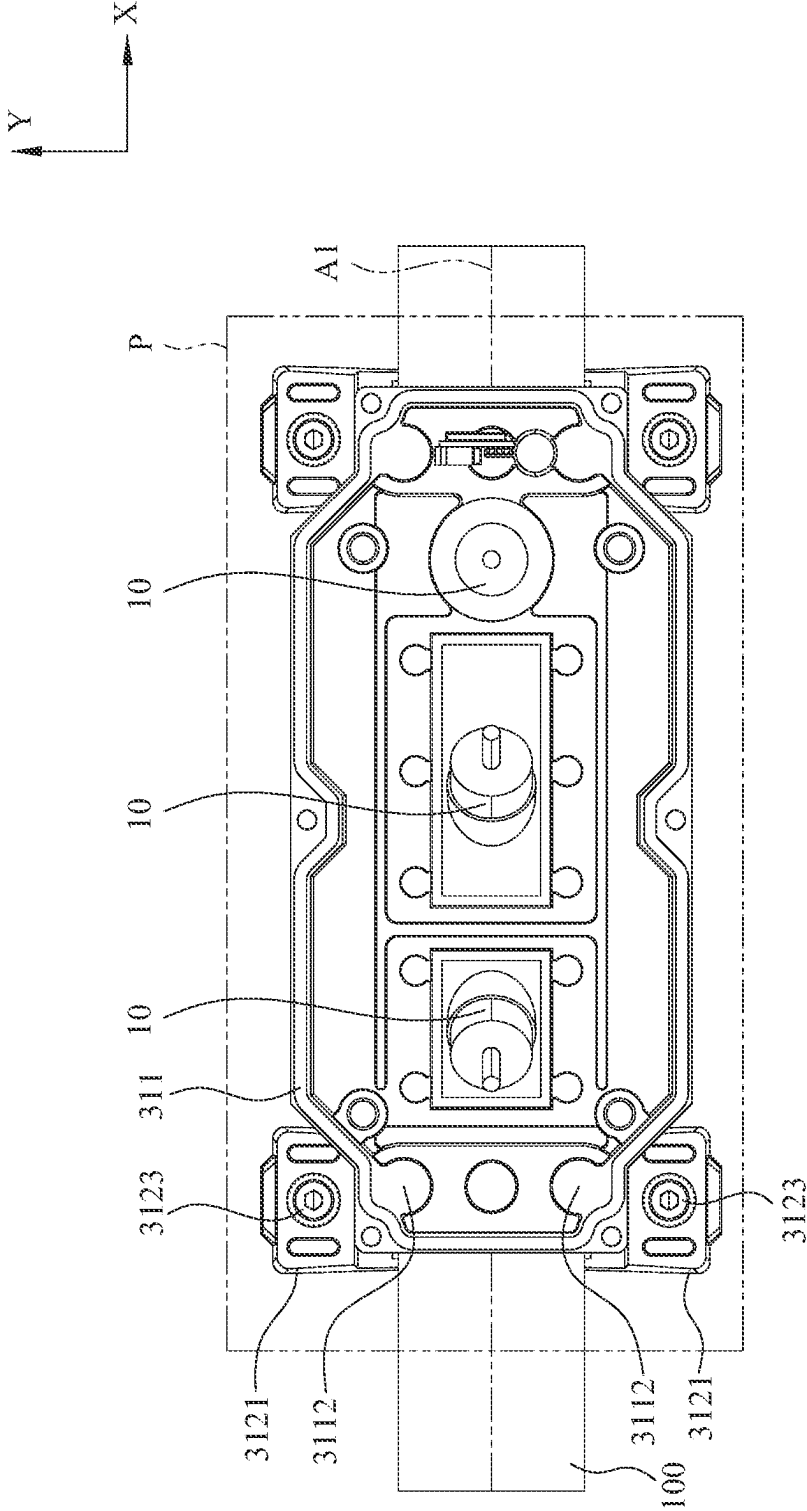


FIG. 3B

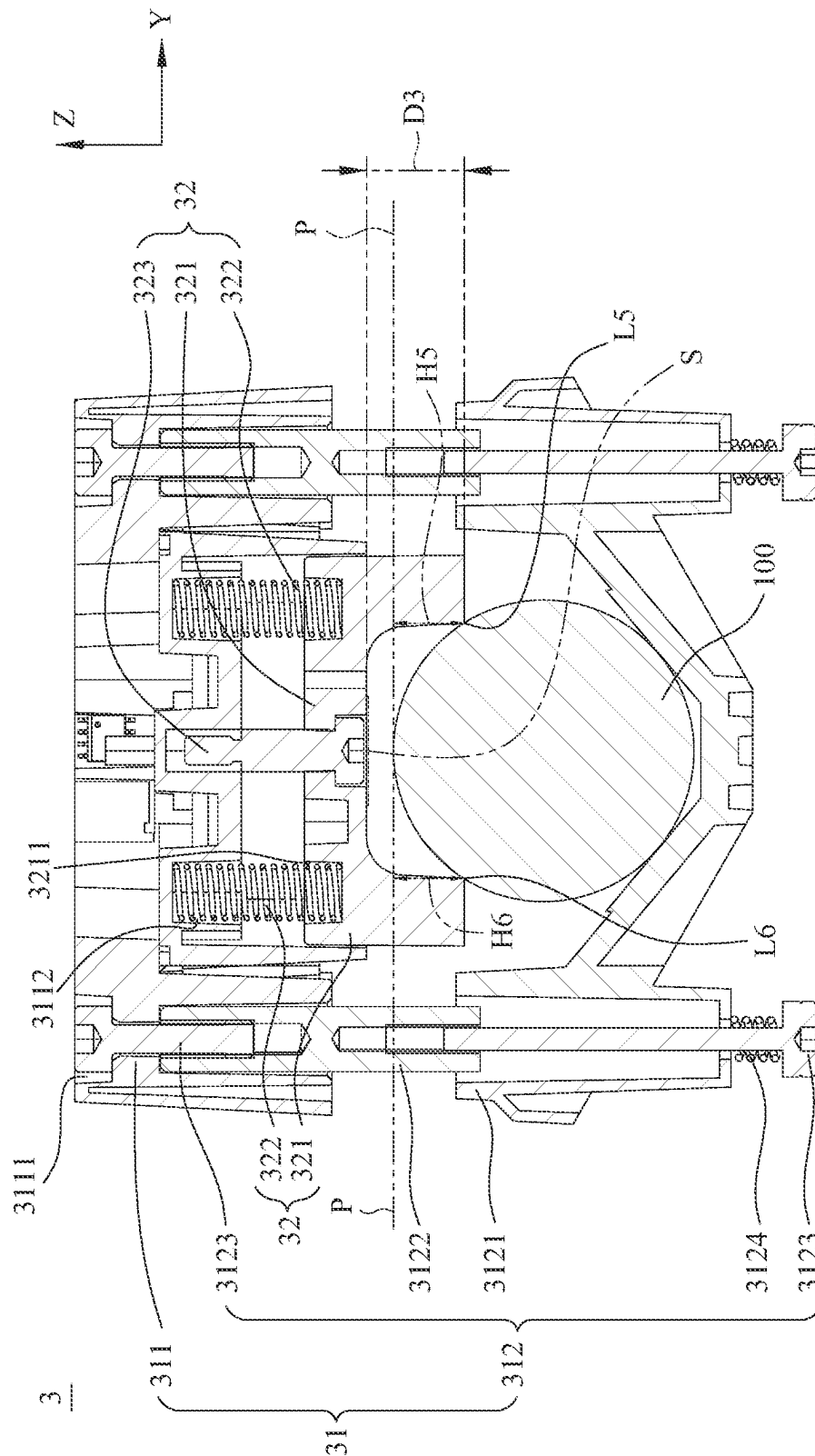


FIG. 30

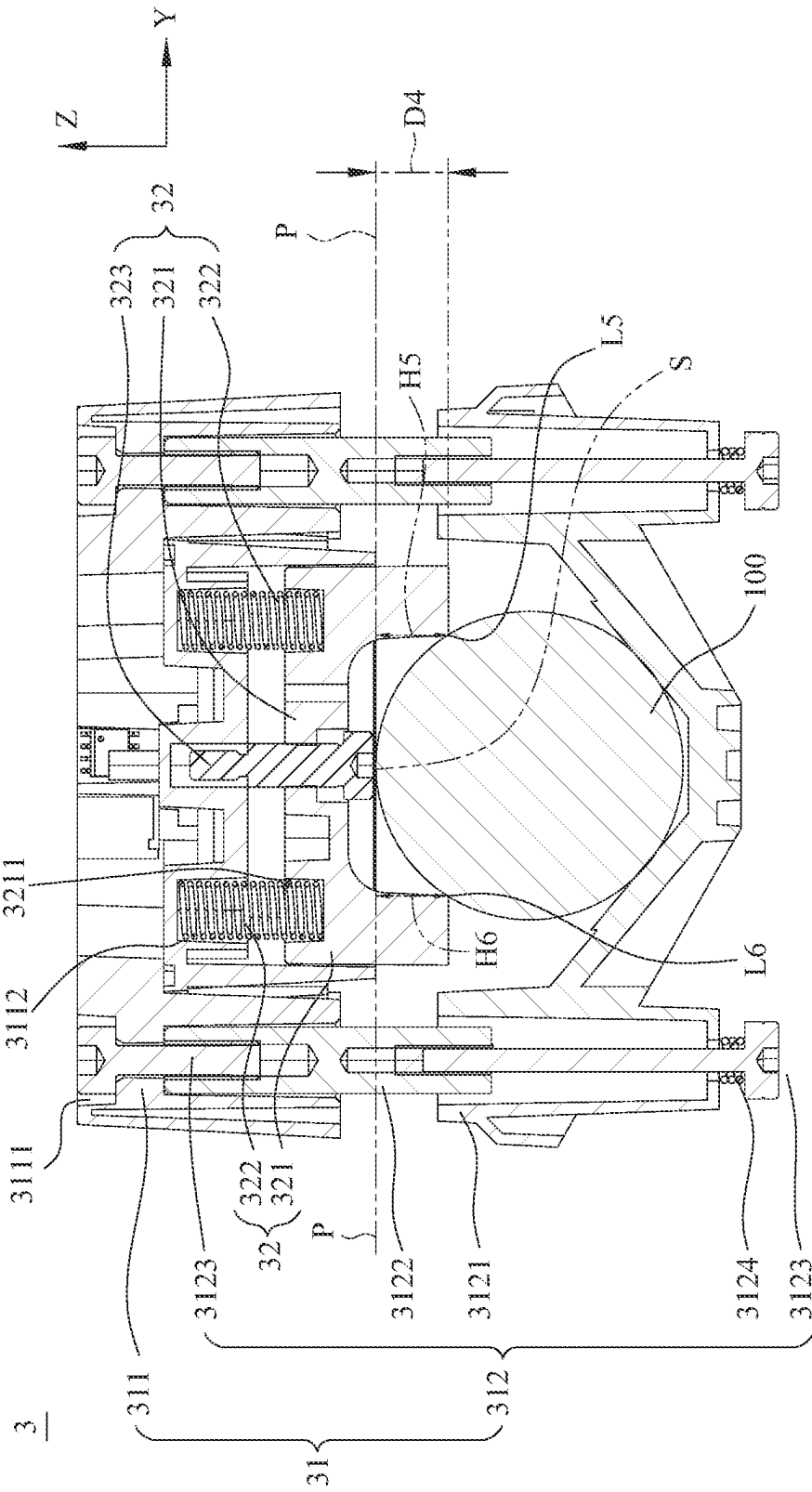


FIG. 3D

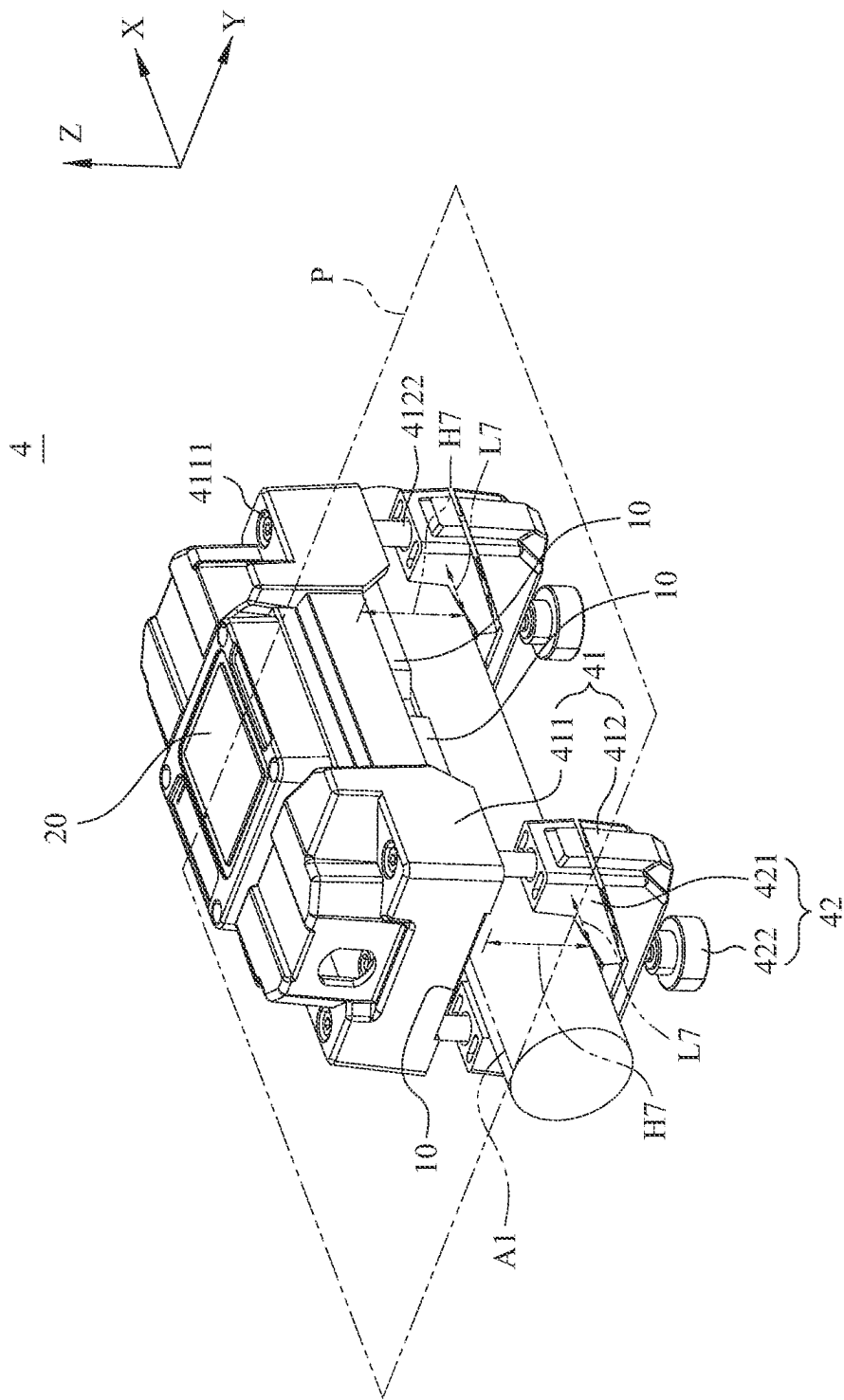


FIG. 4A

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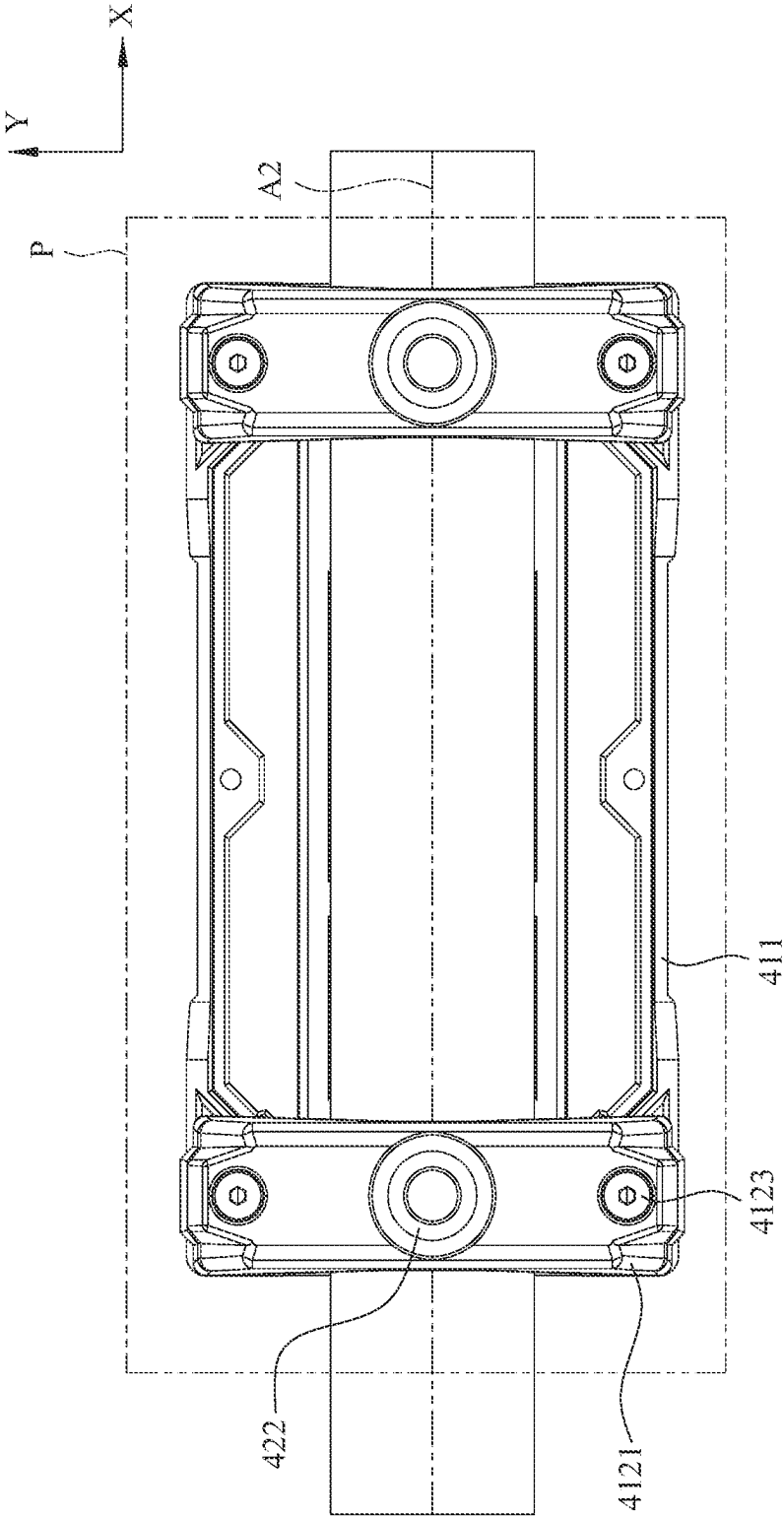


FIG. 4B

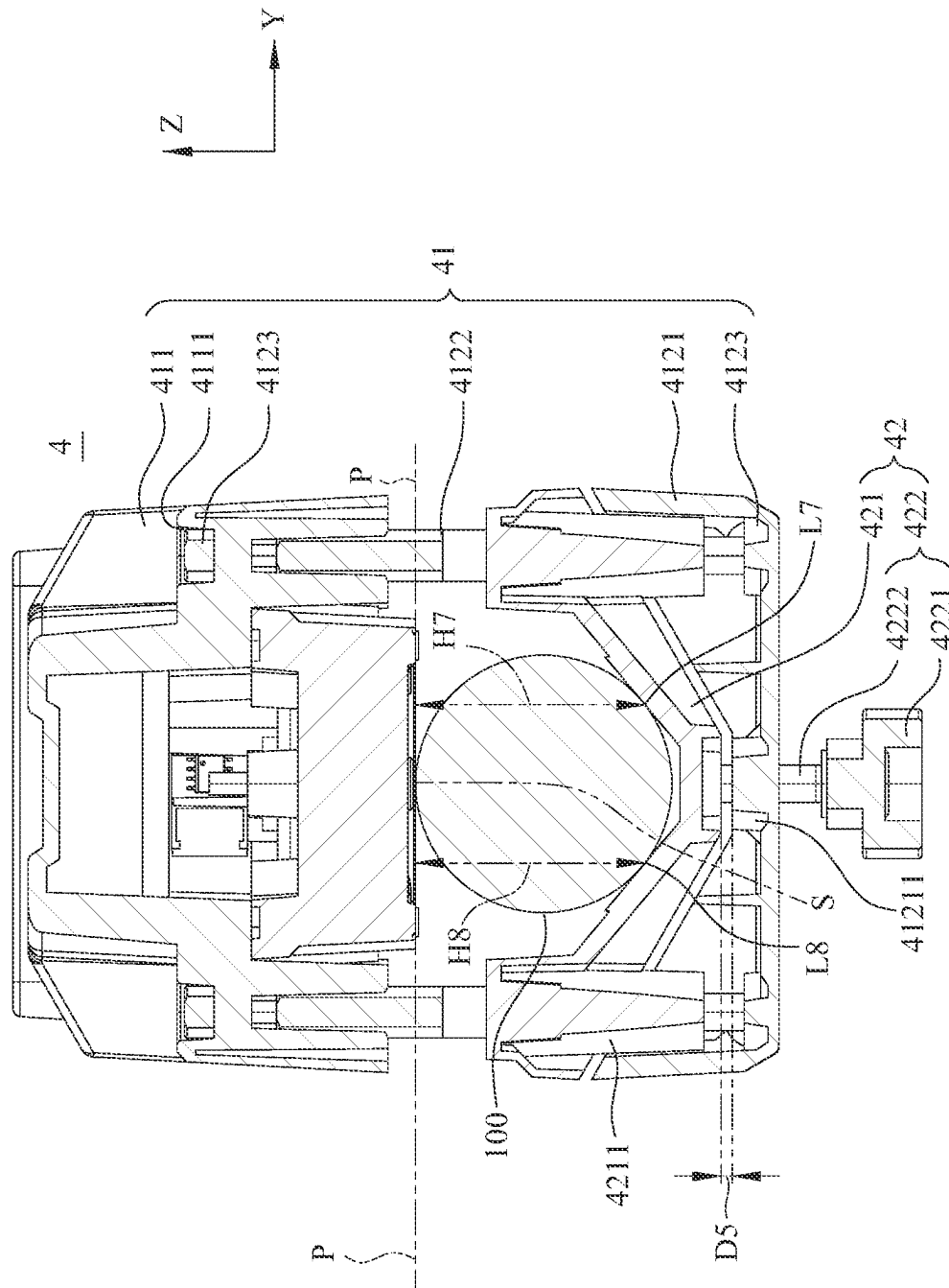


FIG. 4C

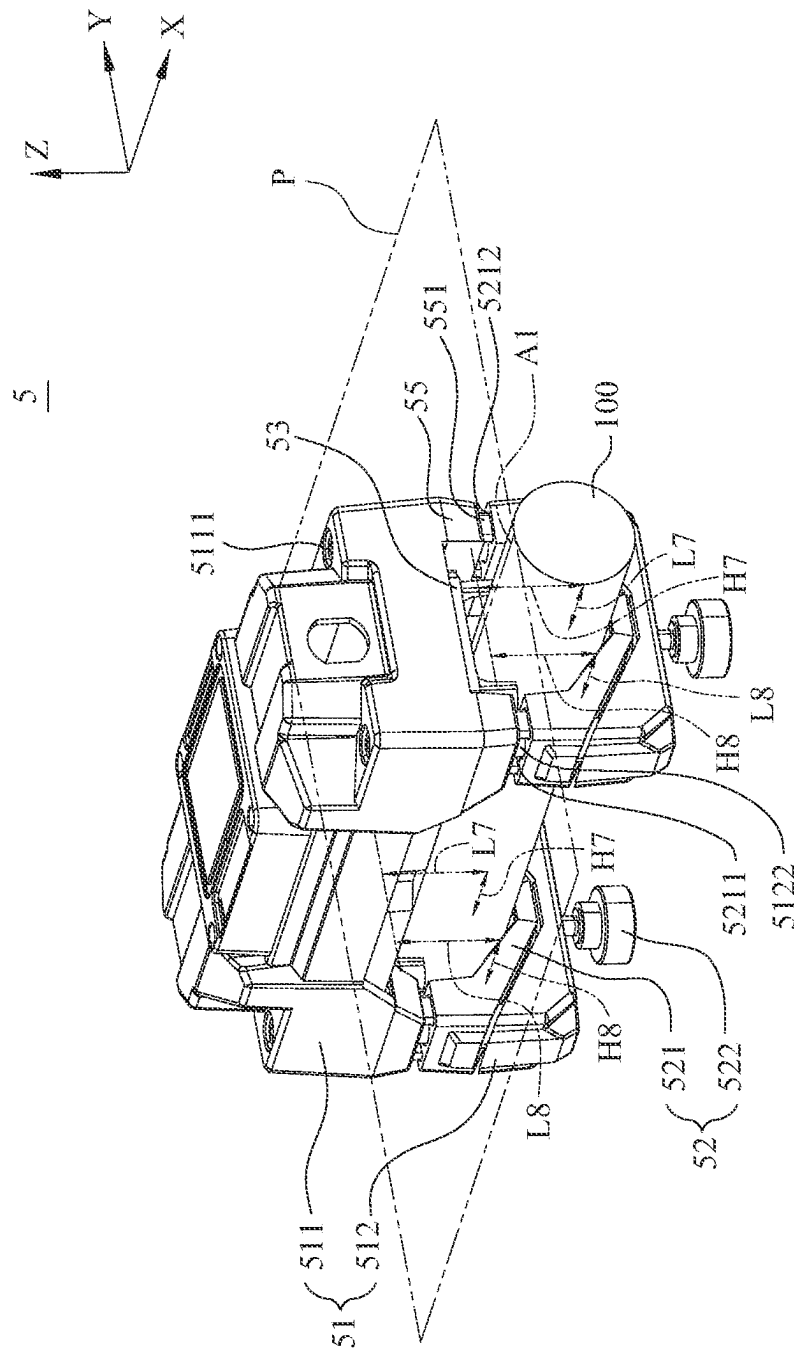


FIG. 5A

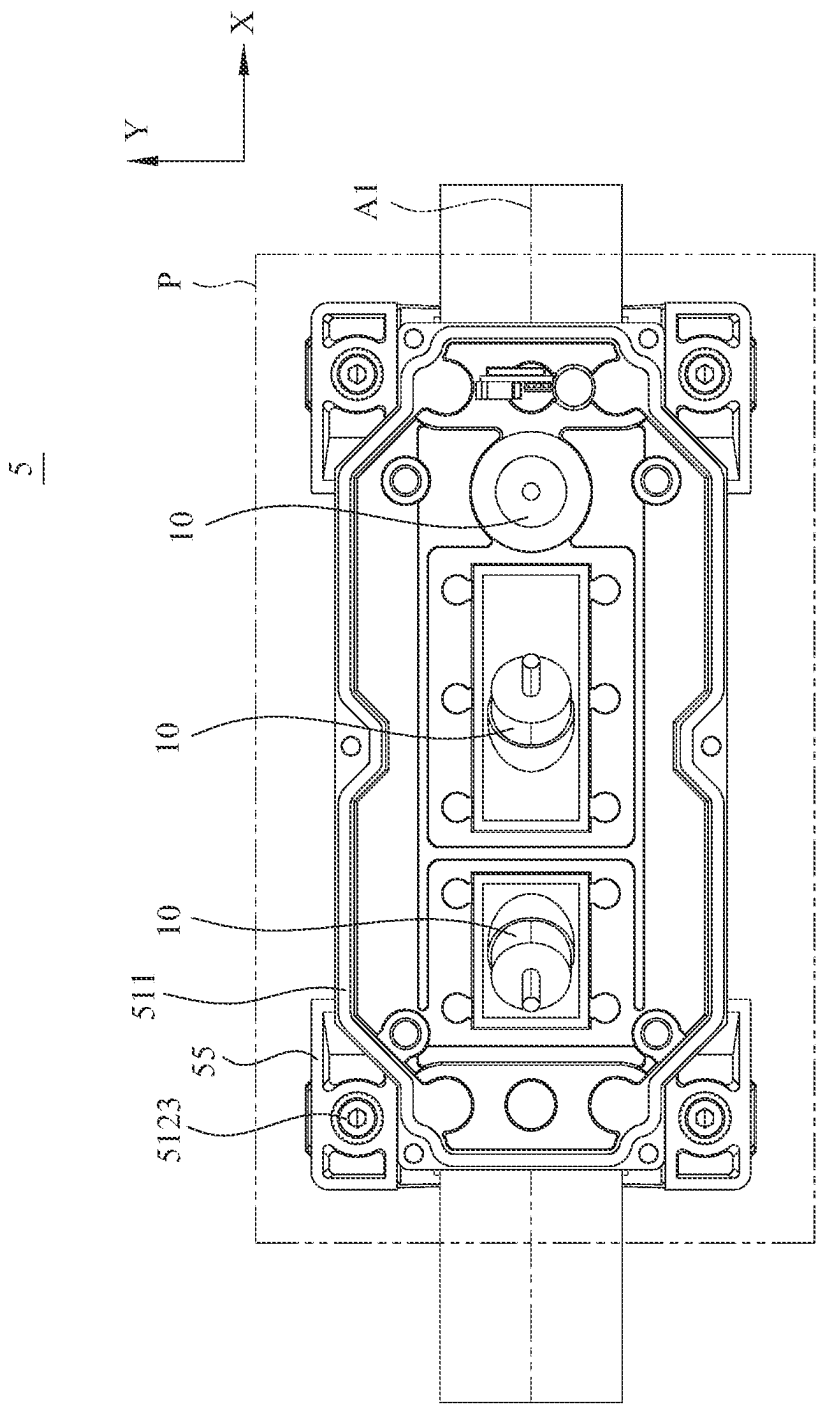
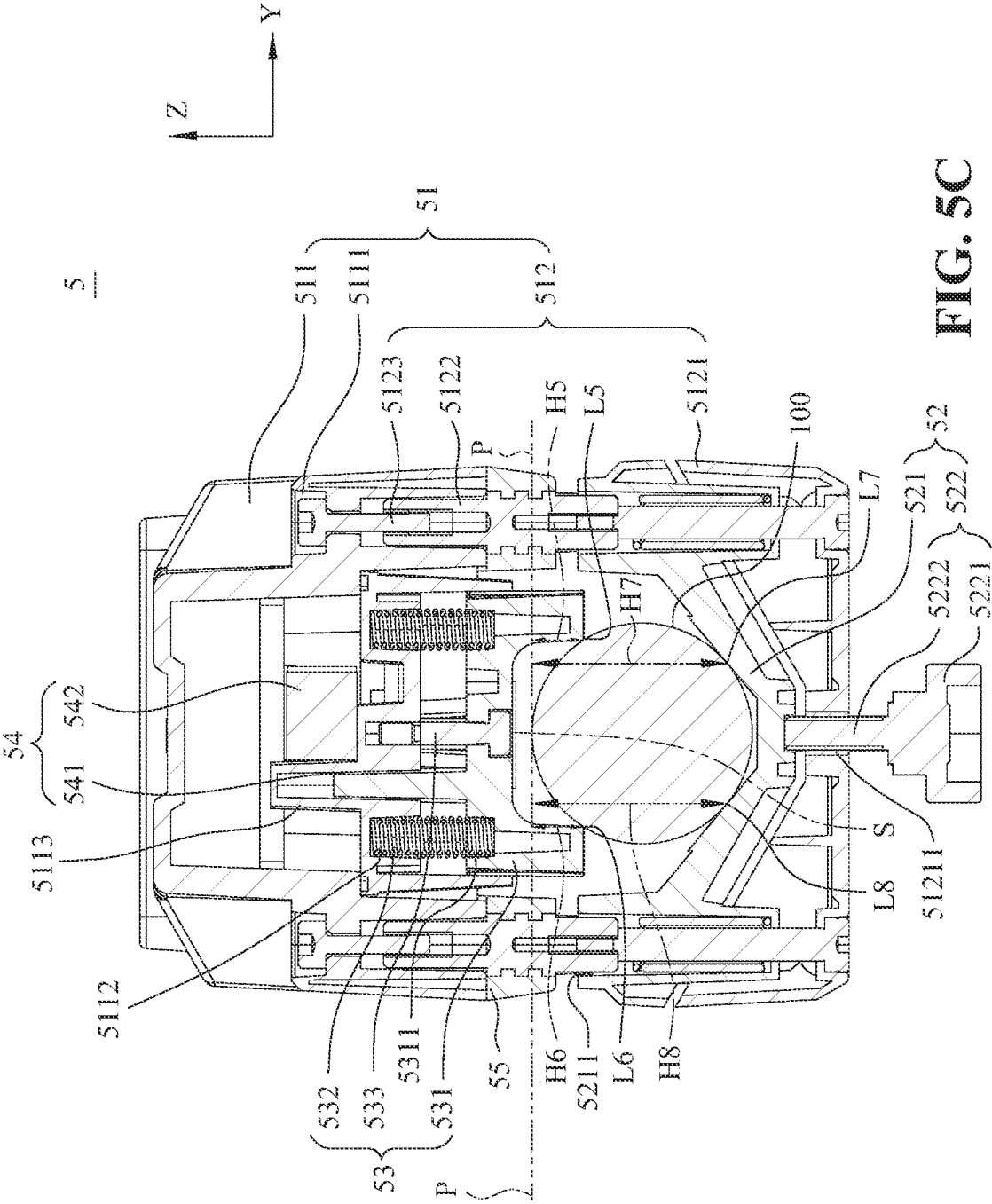
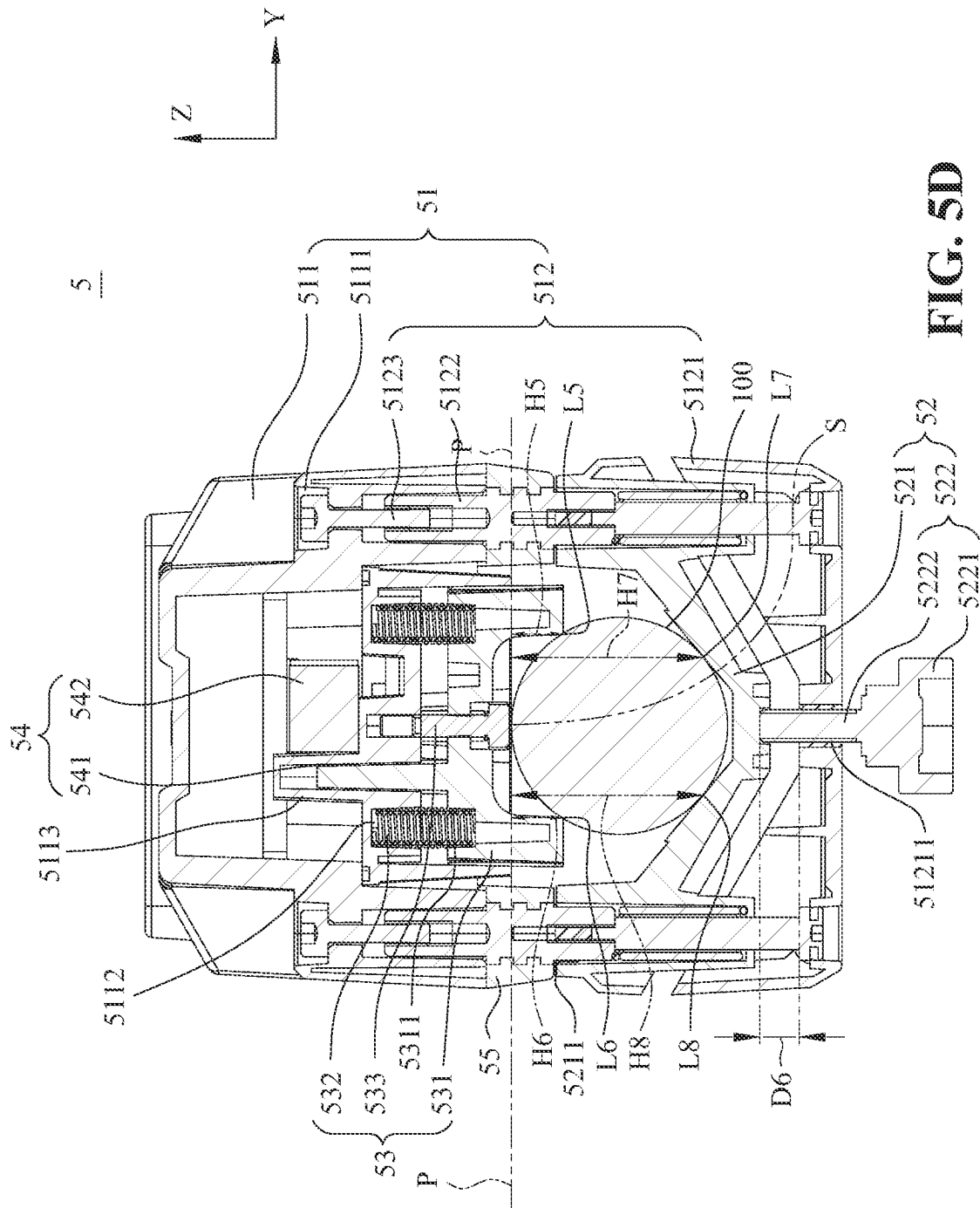


FIG. 5B





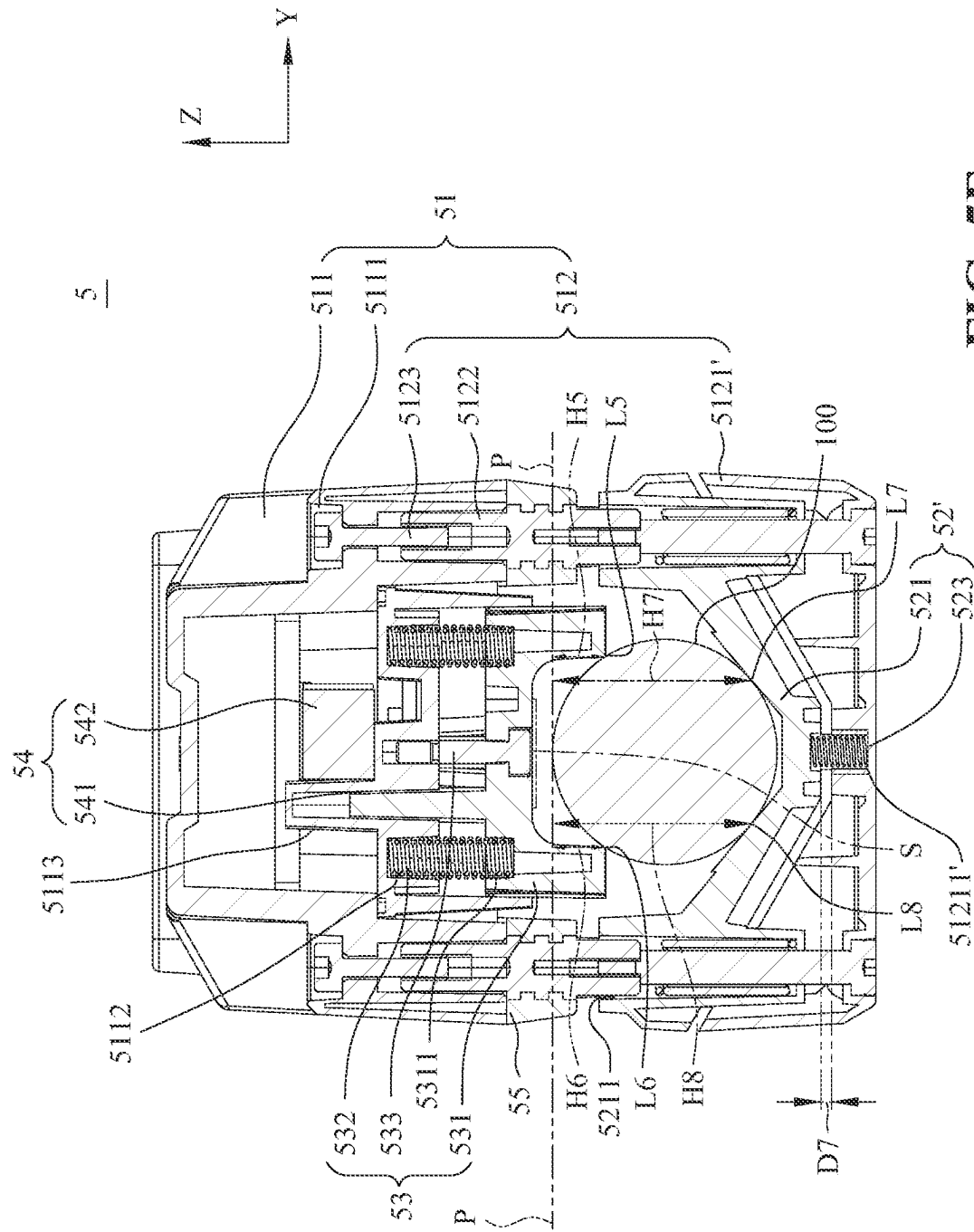


FIG. 5

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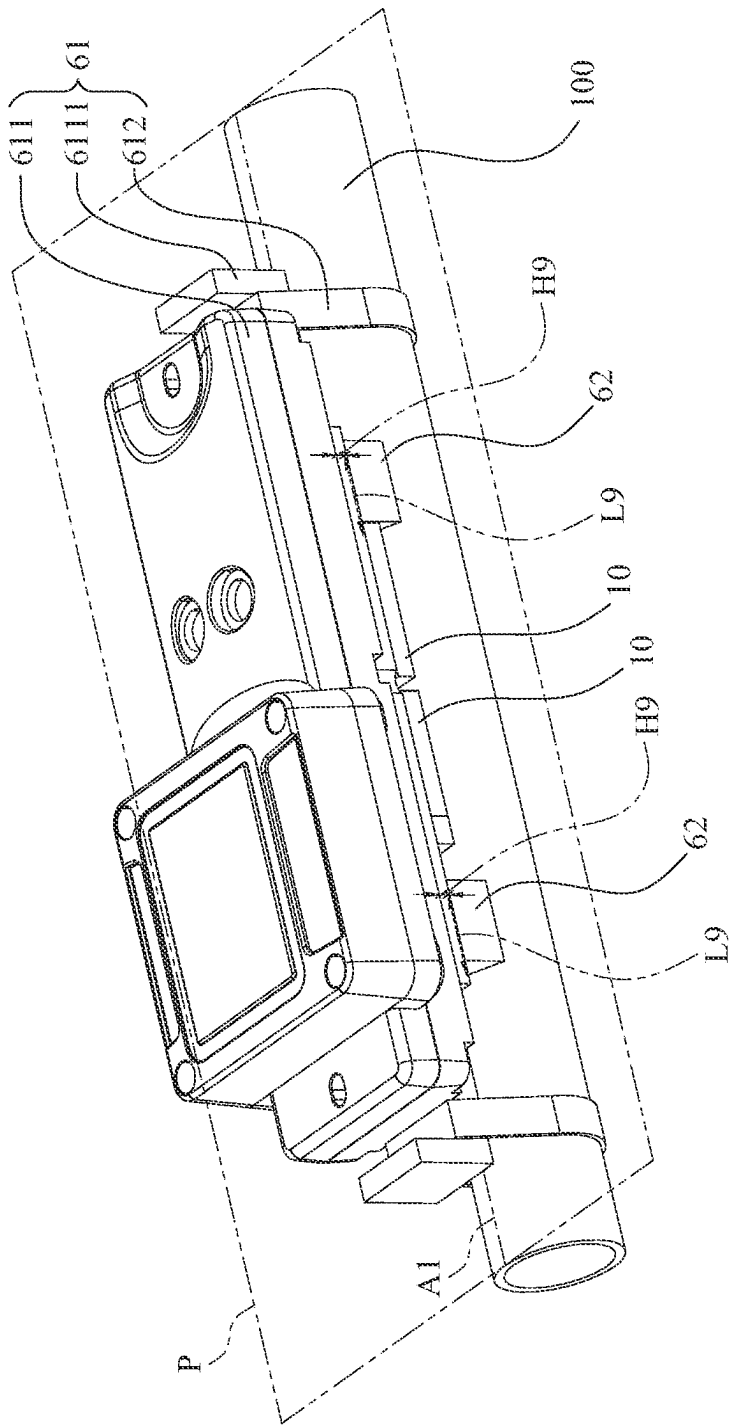


FIG. 6A

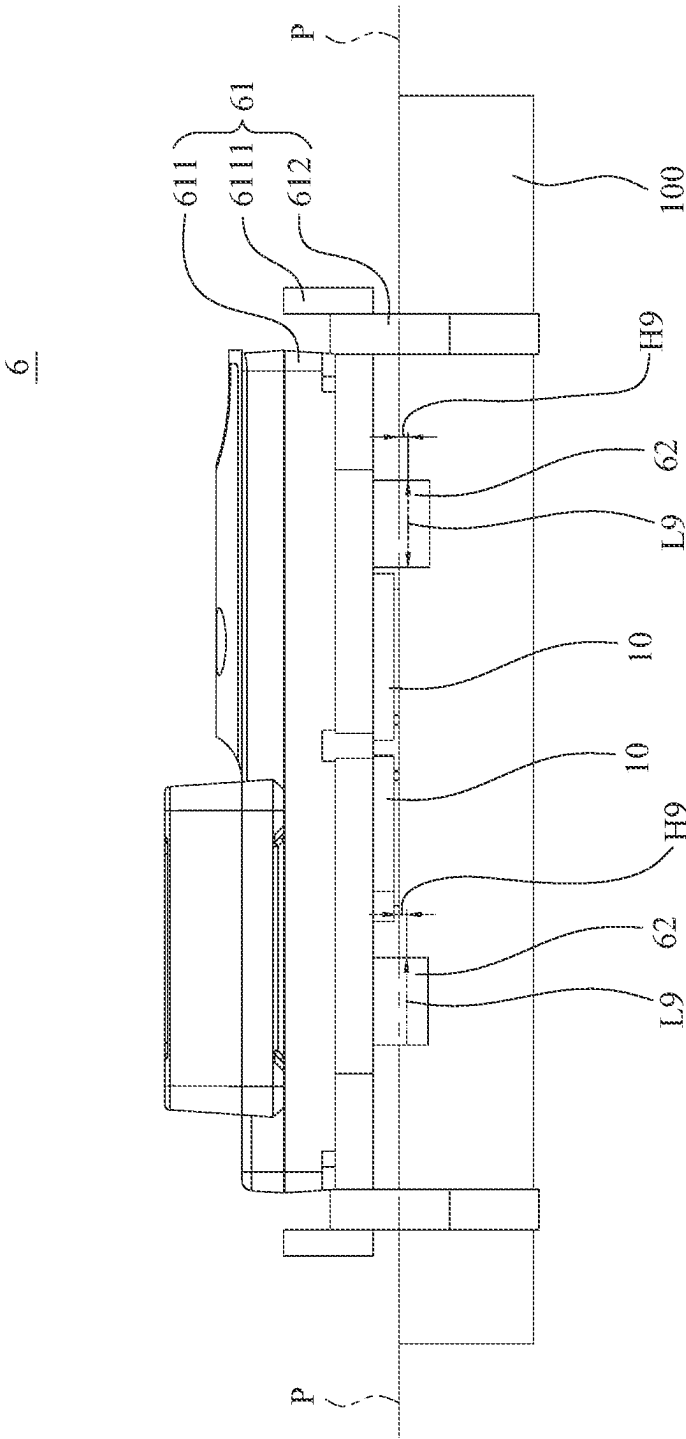


FIG. 6B

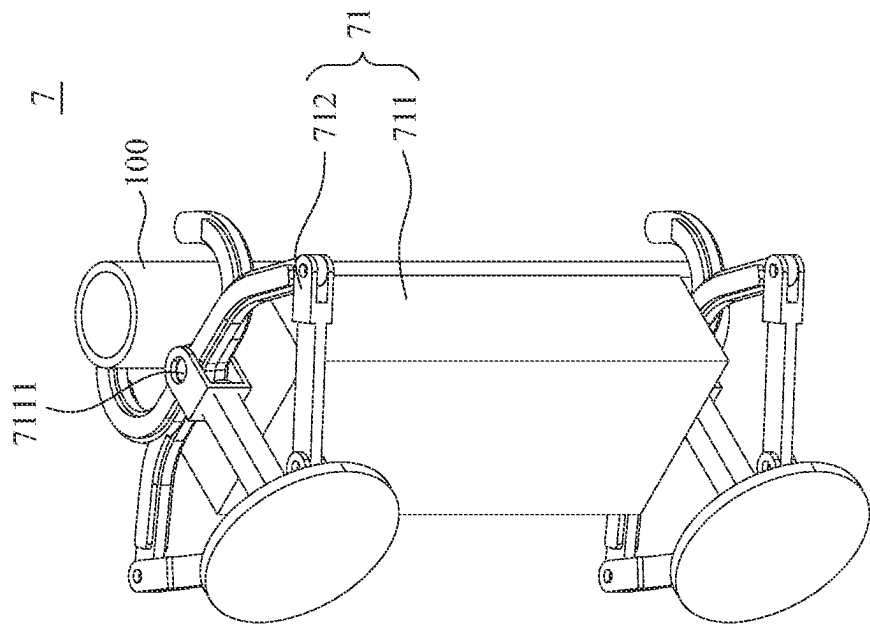


FIG. 7A

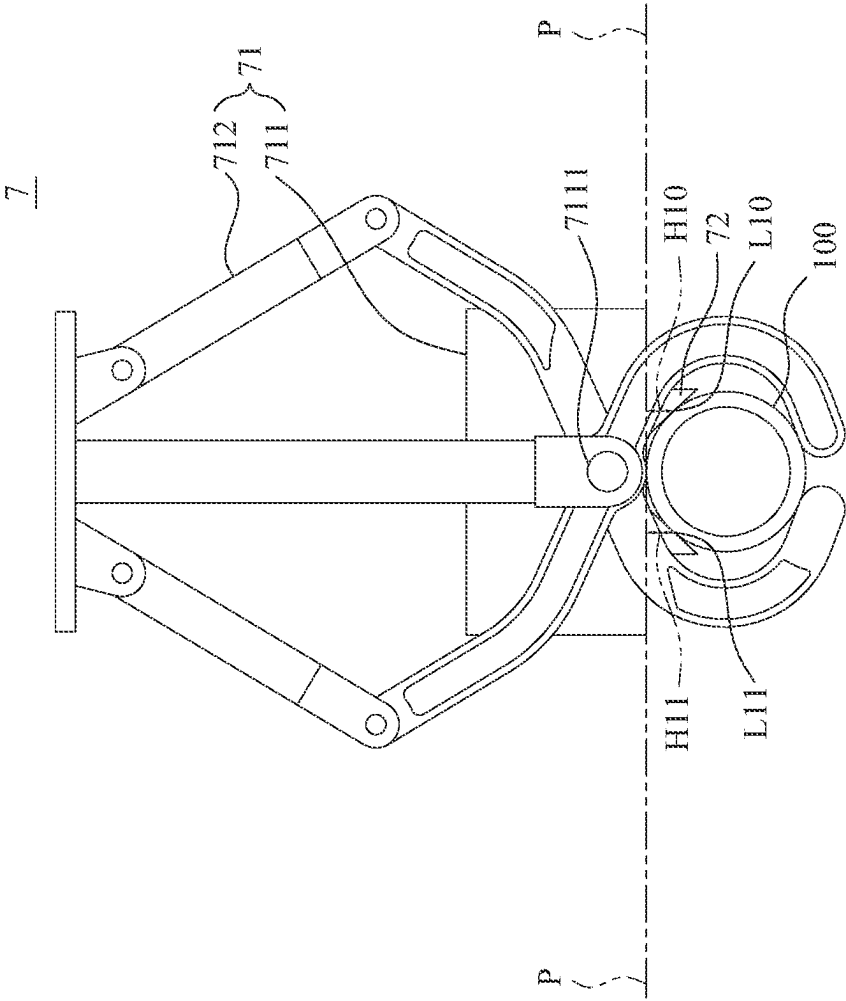


FIG. 7B

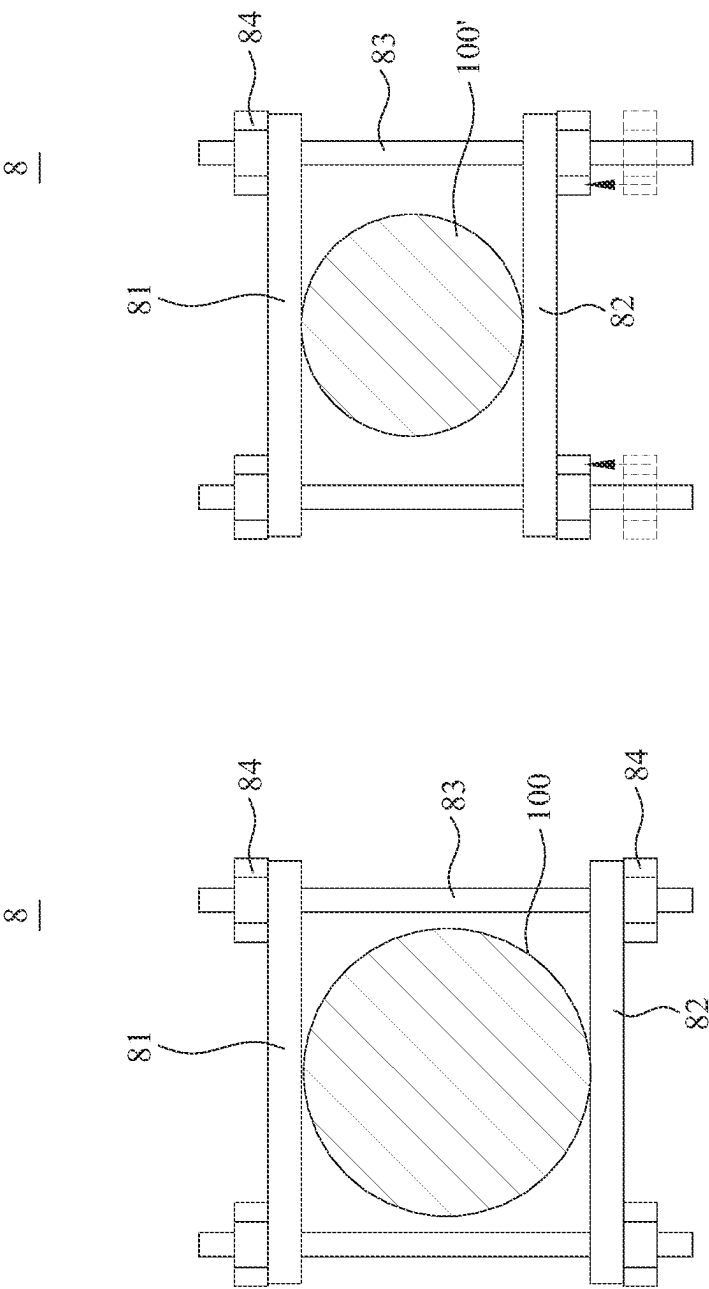


FIG. 8A
(PRIOR ART)

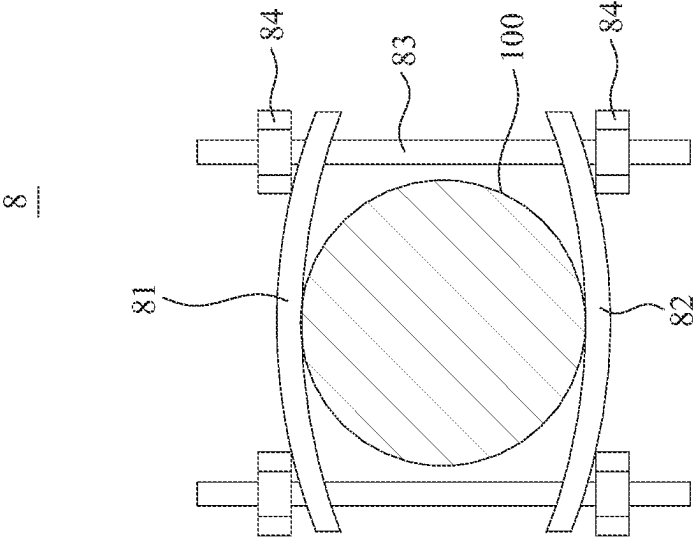


FIG. 8B
(PRIOR ART)

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CLAMP-ON TYPE PIPELINE MEASURING DEVICES AND FIXING STRUCTURES OF THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority of Taiwanese patent application No. 111119735, filed on May 26, 2022, which is incorporated herewith by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates generally to a fixing structure of a measuring device, and more particularly, to a clamp-on type pipeline measuring device and a fixing structure of the same.

2. The Prior Arts

Full-time measuring state of pipelines and materials transported therein is necessary for many industries. Distinguished by installation methods, pipeline measuring devices include embedded type and clamp-on type. The embedded type pipeline measuring device (such as, electromagnetic flow meter) provides higher accuracy, but its installation time period is limited and its installation cost is high. On the other hand, the clamp-on type pipeline measuring device (such as, time-difference ultrasonic flow meter) has lower accuracy, but it has advantages of planning/installation at any time and lower installation cost, which have become the main trend of pipeline measuring technology.

FIG. 8A and FIG. 8B are radial schematic views of the clamps of the conventional clamp-on type pipeline measuring device. As shown in FIG. 8A, the clamp 8 of the conventional clamp-on type pipeline measuring device includes an upper clamp 81, a lower clamp 82, a screw 83, and a nut 84. The upper clamp 81 and the lower clamp 82 are fixed by the screw 83 with the nut 84 on the opposite sides of the outer surface of a pipeline 100. A carrier (not shown) equipped with sensing elements, such as ultrasonic probes, temperature sensors, and vibration sensors, is fixed to the upper clamp 81 or both the upper and lower clamps 81, 82 with screws. Since the same nominal pipe diameter, for example, 1½" pipe, has different reference outer diameters, for example, 48 to 50 mm, and tolerances, for example, ±0.15 to 0.4 mm, in different pipeline standards, when the clamp 8 clamps the pipeline, the radial dimension of the clamp 8 must reserve a residual. When the clamp-on type pipeline measuring device is installed on pipelines of different outer diameters, it is necessary to use screws 83 and nuts 84 to reduce the residual of the clamp 8, as shown by the dotted line in FIG. 8A, so as to fasten the clamp 8 and the pipelines 100, 100'. Then, the carrier and the clamp 8 are tightened with screws, so that a sensing surface of the sensing element can be attached to the outer surface of the pipeline 100, 100' of different outer diameters. As shown in FIG. 8B, in the process of fixing the clamp and the carrier, if the force applied by the screw is uneven or excessive, or the centerline of the carrier fixing mechanism of the clamp 8 and the axis of the pipeline 100, 100' are not parallel or not on the same plane, it is easy to cause the clamp 8 to be deflected or warped (e.g., one or both sides thereof are recessed along the pipeline direction), so that the sensing element in the carrier cannot be tightly attached to the outer

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surface of the pipeline or alignment of the sensing element is not accurate. In the process of fixing the carrier, due to the huge lever torque of the screw, if the force applied to the screw is uneven or excessive, it is also easy to cause the carrier to be deflected or warped. The poor installation of the clamp 8 and the carrier not only generates incorrect measuring data, but also damages the clamp-on type pipeline measuring device.

Even if the sensing element is not correctly installed on the outer surface of the pipeline, the existing clamp-on type pipeline measuring device usually still generates sensing signals and measuring data, so that users may not be able to detect errors in installation and measurement, which not only fails to achieve the purpose of full-time measuring, but also affects production costs and factory safety. Therefore, how to improve the convenience and correctness of installing the clamp-on type pipeline measuring device is the main objective of the present invention.

SUMMARY OF THE INVENTION

In order to achieve the foregoing objective, the present invention provides a fixing structure of a clamp-on type pipeline measuring device, including: a clamp and an aligning member; the clamp is detachably fixed on an outside surface of a pipeline along an axis of the outer surface of the pipeline, the aligning member is disposed in the clamp, wherein a plane tangent to the outer surface of the pipeline on the axis is defined as an installation datum plane. During detaching and fixing the clamp, the aligning member moves along a normal direction of the installation datum plane; after the clamp is fixed to the outer surface of the pipeline, the aligning member contacts at least two line segments on the outer surface of the pipeline, and distances between the two line segments and the installation datum plane are the same.

In an embodiment, the clamp includes a carrier and a buckle, the carrier is set on the installation datum plane, and the buckle surrounds the outer surface of the pipeline and connects to the carrier.

In an embodiment, the aligning member is arranged in the carrier, and a distance that the aligning member moves along the normal direction of the installation datum plane is proportional to a size of the outer diameter of the pipeline.

In an embodiment, the aligning member is arranged in the buckle, and a distance that the aligning member moves along the normal direction of the installation datum plane is inversely proportional to a size of the outer diameter of the pipeline.

In an embodiment, the fixing structure of the clamp-on type pipeline measuring device further includes an outer diameter measuring unit, which is arranged on the carrier, and the outer diameter measuring unit moves along the normal direction of the installation datum plane during the process of disassembling and fixing the buckle, and a distance that the outer diameter measuring unit moves along the normal direction of the installation datum plane is proportional to a size of the outer diameter of the pipeline.

In an embodiment, the aligning member includes a slider, the slider has two protruding arms, and a surface between the two arms is in the shape of a single angle, a polygon, a plane or an arc.

In an embodiment, the aligning member further includes an elastic member, and two ends of the elastic member are respectively embedded in the clamp and the slider, so as to generate an elastic force for moving the aligning member along the normal direction of the installation datum plane.

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The present invention provides a clamp-on type pipeline measuring device, including a fixing structure and a sensing element, the fixing structure includes a clamp and an aligning member, wherein the clamp is detachably fixed on an outer surface of a pipeline, the aligning member is disposed in the clamp; the sensing element is arranged in the clamp and has a sensing surface; when the outer surface of the pipeline contacts the sensing surface, the clamp is fixed on the outer surface of the pipeline, the aligning member contacts at least two line segments of the outer surface of the pipeline, and the distances between the two line segments and the sensing surface are the same.

In an embodiment, the clamp includes a carrier and a buckle, and the aligning member is arranged in the carrier or the buckle.

In an embodiment, the sensing element is at least one of an ultrasonic probe, a temperature sensor, and a vibration sensor.

The fixing structure of the clamp-on type pipeline measuring device of the present invention includes a clamp and an aligning member, and the aligning member can be applied to pipelines with different pipe diameters, especially those with the same nominal pipe diameter and different pipeline specifications. In the process of fixing the clamp and the pipeline, the aligning member moves along the normal direction of the tangent plane passing through the axis of the outer surface of the pipeline, so that the center line of the clamp can be aligned with the axis of the outer surface of the pipeline. After fixing the clamp on the outer surface of the pipeline, the aligning member abuts at least two line segments equidistant from the outer surface of the pipeline and the installation datum plane, which can keep the clamp and the sensing element in the correct installation position, so that the sensing surface of the sensing element can be set on the tangent plane to the axis of the outer surface of the pipeline. The clamp-on type pipeline measuring device of the present invention reduces the use of screws to adjust the residual between the clamp and the pipeline, and effectively prevents the clamp, pipeline and sensing elements from shifting, warping, deforming and damage due to improper installation, improves the convenience of installation, and ensures the correctness of measurement.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention will be apparent to those skilled in the art by reading the following detailed description of a preferred embodiment thereof, with reference to the attached drawings, in which:

FIG. 1A is a side view of the fixing structure of the clamp-on type measuring device of a first embodiment of the present invention; FIG. 1B is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 1A before fixed on the pipeline; FIG. 1C is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 1A after fixed on the pipeline; and FIG. 1D is a radial schematic view of other aligning members of the first embodiment of the present invention;

FIG. 2A is a side view of the fixing structure of the clamp-on type measuring device of a second embodiment of the present invention; FIG. 2B is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 2A before fixed on the pipeline; FIG. 2C is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 2A after

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fixed on the pipeline; and FIG. 2D is a radial schematic view of other aligning members of the second embodiment of the present invention;

FIG. 3A is a side view of the fixing structure of the clamp-on type measuring device of a third embodiment of the present invention; FIG. 3B is a top cross-sectional view of the other aligning members shown in FIG. 3A; FIG. 3C is radial cross-section view of the fixing structure of the clamp-on type measuring device shown in FIG. 3A before fixed on the pipeline; and FIG. 3D is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 3A after fixed on the pipeline;

FIG. 4A is a side view of the fixing structure of the clamp-on type measuring device of a fourth embodiment of the present invention; FIG. 4B is a bottom view of the other aligning members shown in FIG. 4A; and FIG. 4C is radial cross-section view of the fixing structure of the clamp-on type measuring device of FIG. 4A;

FIG. 5A is a side view of the fixing structure of the clamp-on type measuring device of a fifth embodiment of the present invention; FIG. 5B is a top view of the other aligning members shown in FIG. 5A; FIG. 5C is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 5A before fixed on the pipeline; FIG. 5D is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 5A after fixed on the pipeline; and FIG. 5E is a radial schematic view of other aligning members of the fifth embodiment of the present invention;

FIG. 6A is a side view of the fixing structure of the clamp-on type measuring device of a sixth embodiment of the present invention; and FIG. 6B is an axial view of the fixing structure of the clamp-on type measuring device shown in FIG. 6A;

FIG. 7A is a side view of the fixing structure of the clamp-on type measuring device of a seventh embodiment of the present invention; and FIG. 7B is a radial view of a fixing structure of the clamp-on type measuring device shown in FIG. 7A; and

FIG. 8A and FIG. 8B are radial views of the fixing structure of the conventional clamp-on type measuring device.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention. Unless the context clearly indicates otherwise, terms herein include singular and plural forms, and the term "and/or" includes any and all combinations of one or more of the associated listed items.

The outer surface of the pipeline is cylindrical, and the sensing surface of the sensing element of the clamp-on type pipeline measuring device is flat. Under the condition that the pipeline and the sensing element are not deformed, the correct installation position of the sensing surface of the clamp-on type pipeline measuring device is the tangent plane of the outer surface of the pipeline, and the sensing surface contacts an axis of the outer surface of the pipeline. FIG. 1A is a side view of the fixing structure of the clamp-on type measuring device of a first embodiment of the present invention; FIG. 1B is radial cross-section view of the fixing structure of the clamp-on type measuring device shown in

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FIG. 1A before fixed on the pipeline; FIG. 1C is radial cross-section view of the fixing structure of the clamp-on type measuring device shown in FIG. 1A after fixed on the pipeline. As shown in FIGS. 1A, 1B and 1C, the fixing structure 1 of the clamp-on type pipeline measuring device includes: a clamp 11 and an aligning member 12, the clamp 11 is detachably fixed on the outer surface of the pipeline 100 along the axis A1 (as X-axis direction shown in FIG. 1A) of the outer surface of the pipeline 100, and the aligning member 12 is disposed in the clamp 11, wherein the plane tangent to the axis A1 with the outer surface of the pipeline 100 is defined as the installation datum plane P (as the X-Y plane shown in FIG. 1A). During the processes of detaching and fixing the clamp 11, the aligning member 12 moves along the normal line of the installation datum plane P (as the Z-axis direction shown in FIG. 1A). After the clamp 11 is fixed on the outer surface of the pipeline 100, the aligning member 12 contacts at least two line segments L1, L2 on the outer surface of the pipeline 100, and the distances H1, H2 between the line segments L1, L2 and the installation datum plane P are the same.

In this embodiment, the clamp 11 includes a carrier 111 and a buckle 112. The carrier 111 can be made of plastic or alloy molding, is roughly rectangular, and can be equipped with a sensing element 10 and a processing circuit (not shown), a power supply (not shown), a display screen 20, etc. The sensing elements 10 can be ultrasonic probes, temperature sensors, vibration sensors, etc. The number of sensing elements 10 can be one or plural. The plurality of sensing elements 10 can be arranged in the carrier 111 along the long side direction of the carrier 111, and the two sides of the long side of the carrier 111 have a buckle seat 1111 and an aligning groove 1112. The buckle 112 includes an arc seat 1121 and a bolt 1122, can surround the outer surface of the pipeline 100, and connect the buckle seat 1111 of the carrier 111. The number of aligning members 12 can be one or plural; for example, a single aligning member 12 is arranged in the center of the long side of the carrier 111; two aligning members 12 can be arranged on the two sides of the long side of the carrier 111. The aligning member 12 includes a slider 121, a guide block 122, and an elastic member 123. A shape of the slider 121 is about "A", one end of the guide block 122 is connected to center of the slider 121, the other end of guide block 122 is embedded in the aligning groove 1112 located in the center of the short side of the carrier 111, and the two ends of the elastic member 123 (such as but not limited to a spring, leaf spring and rubber) are respectively connected to an inner wall of the aligning groove 1112 and the guide block 122. The elastic member 123 can generate an elastic force to move the slider 121 and a tensile force to prevent the slider 121 from moving out of the carrier 111. Depending on the outer diameter range of the pipeline 100, the aligning member 12 may move a distance of 0.1 mm to 10 cm in the aligning groove 1112 towards or away from the carrier 111.

The process of installing the clamp 11 on the outer surface of the pipeline 100 includes: placing the pipeline 100 on the arc seat 1121, placing the long side of the carrier 111 on the outer surface of the arc seat 1121 along the axial direction of the pipeline 100, so as to make the slider 121 of the aligning member 12 abut against the four line segments on the opposite sides of the axis A1 of the outer surface of the pipeline 100; screwing the bolt 1122 through the buckle seats 1111 on both sides of the carrier 111 into the screw hole of the arc seat 1121, to make the carrier 111 and the arc seat 1121 move towards each other to clamp the opposite sides of the outer surfaces of the pipeline 100, and concurrently to

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make the outer surface of the pipeline 100 push the aligning member 12 gradually moving toward the carrier 111 along the normal line vertical to the installation datum plane P (as the Z-axis direction shown in FIG. 1C). Since the elastic member 123 of the aligning member 12 is pressed to generate the elastic force that makes the slider 121 abut against the line segments L1 and L2 on the opposite sides of the outer surface of the pipeline 100 and the distance between the two line segments L1, L2 and the installation datum plane P be kept the same during the moving of the aligning member 12, so that the center line (the line connecting the center points of the two short sides of the bottom surface of the carrier) of the bottom surface of the carrier 111 (i.e., the surface of the carrier facing the pipeline is defined as the bottom surface of the carrier) can be aligned with the axis A1 of the outer surface of the pipeline 100 without deviation. When the bolt 1122 is screwed into the arc seat 1121 until the axis A1 of the outer surface of the pipeline 100 contacts the sensing surface S of the sensing element 10, and the clamp 11 will not move along the axis A1 of the pipeline 100 or rotate along the circumference of the pipeline 100, it means that the clamp 11 has been correctly fixed on the outer surface of the pipeline 100; that is, the centerline of the bottom surface of the carrier 111 is aligned with the axis A1 of the outer surface of the pipeline 100 and the sensing surface S of the sensing element 10 is set on the installation datum plane P.

The main function of the aligning member is to move with the clamp and abut against at least two line segments on the outer surface of the pipeline during the processes of detaching and fixing. Therefore, the slider has two protruding arms, and the shape thereof is not limited to the A shape illustrated in the figure. The surface of the recess between the two arms can be single angle, polygon, plane or arc. FIG. 1D is a schematic radial view of other aligning members of the first embodiment. As shown in FIG. 1D, the fixing structure 1', 1" of the clamp-on type pipeline measuring device includes: a clamp 11 and aligning members 12', 12", and the slider 121' of the aligning member 12' is roughly arched in shape, the slider 121" of the aligning member 12" is roughly in the shape of an arch bridge, and the aligning members 12', 12" have the same functions and effects as the aforesaid aligning member 12.

FIG. 2A is a side view of the fixing structure of the clamp-on type measuring device of a second embodiment of the present invention; FIG. 2B is radial cross-section view of the fixing structure of the clamp-on type measuring device shown in FIG. 2A before fixed on the pipeline; FIG. 2C is radial cross-section view of the fixing structure of the clamp-on type measuring device shown in FIG. 2A after fixed on the pipeline. As shown in FIGS. 2A, 2B and 2C, the fixing structure 2 of the clamp-on type pipeline measuring device includes: a clamp 21 and an aligning member 22, the clamp 21 is detachably fixed on the outer surface of the pipeline 100 along the axis A1 of the outer surface of the pipeline 100 (as X-axis direction shown in FIG. 2A), and the aligning member 22 is disposed in the clamp 21, wherein the plane tangent to the axis A1 with the outer surface of the pipeline 100 is defined as the installation datum plane P (as X-Y plane shown in FIG. 2A). After the clamp 21 is fixed on the outer surface of the pipeline 100, the aligning member 22 contacts at least two line segments L3, L4 on the outer surface of the pipeline 100, and the distances H3, H4 between the line segments L3, L4 and the installation datum plane P are the same.

In this embodiment, the clamp 21 includes a carrier 211 and a buckle 212, the carrier 211 can be made of plastic or

alloy molding, the carrier **211** is roughly rectangular, and can be installed with the sensing element **10**, the processing circuit (not shown), power supply (not shown), display screen (not shown), etc. The two sides of the long side of the carrier **211** have a buckle seat **2111**, an aligning groove **2112**, and a slot **2113**, and the buckle **212** includes a U-shaped screws and nuts that can surround the outer surface of the pipeline **100** and connect the two sides of the long side of the carrier **211**. Two aligning members **22** are arranged on the two sides of the long side of the carrier **211**, and the aligning members **22** include sliders **221**, elastic members **222**, and stop blocks **223**. The slider **221** is roughly in the shape of A, the center of the A shape is aligned with the center point of the short side of the carrier **211**, and the two arms of the A shape form a receiving groove **2211**. The two ends of the elastic member **222** are respectively embedded in the aligning groove **2112** and the receiving groove **2211**. The elastic member **222** (such as, but not limited to spring, leaf spring, and rubber) can generate the elastic force to move the slider **221** and the resistance to keep the slider **221** in symmetrical contact with the two sides of the outer surface of the pipeline **100**. A bottom end **2231** of the stop block **223** is fixed to the slot **2113** of the carrier **211**, a top end **2232** of the stop block **223** is inserted into a chute **2212** in the center of the slider **221**, and the surface of the top end **2232** is slightly recessed than the sensing surface S of the sensing element **10**. When the slider **221** moves away from the carrier **211**, the stop block **223** can prevent the aligning member **22** from moving out of the carrier **211**. As shown in FIGS. 2B and 2C, when the clamp **21** is not fixed on the pipeline, the slider **221** protrudes from the bottom surface of the carrier **211** by a length D1; after the clamp **21** is fixed on the outer surface of the pipeline **100**, the slider **211** protrudes from the bottom surface of the carrier **211** by a length D2; the distance (D1-D2) that the slider **221** moves toward or away from the carrier **211** along the normal direction of the installation datum plane P is between 0.1 mm and 10 cm.

The stop block of the aligning member is not limited to the above-mentioned forms. FIG. 2D is a radial schematic diagram of other aligning members according to the second embodiment of the present invention. As shown in FIG. 2D, the aligning member **22'** includes a slider **221**, an elastic member **222**, and a stop block **223'**, and the inner wall of the carrier **211** for accommodating the aligning member **22'** has a guide groove **2114**, and the stop block **223'** forms at the two side walls of the slider **221** and is inserted into the guide groove **2114**. When the slider **221** moves away from the carrier **211**, the stop block **223'** can also stop the aligning member **22'** from moving out of the carrier **211**.

The process of installing the clamp **21** on the outer surface of the pipeline **100** includes: placing the long side of the carrier **211** on the outer surface of the pipeline **100** along the axial direction of the pipeline **100**, so that the slider **221** of the aligning member **22** abuts the four line segments on the opposite sides of the axis A1 of the outer surface of the pipeline **100**; using the U-shaped screw of the buckle **212** to surround the outer surface of the pipeline **100** and passes through the buckle seat **2111** of the carrier **211**, and screwing the nut into the U-shaped screw to make the carrier **211** and the U-shaped screw move on opposite directions to clamp the outer surfaces of the two opposite sides of the pipeline **100**, and concurrently to drive the outer surface of the pipeline **100** to push the aligning member **22** to gradually move toward the carrier **211** along the normal direction perpendicular to the installation datum plane P (as Z-axis direction shown in FIG. 2C). Since the slider **221** of the aligning member **22** abuts against the line segments L3 and

L4 on both sides of the axis A1 of the outer surface of the pipeline **100**, and keeps the same distance from the installation datum plane P during the moving of the aligning member **22**, so that the center line of the bottom surface of the carrier **211** can be aligned with the axis A1 of the outer surface of the pipeline **100** without deviation. When the nut is screwed onto the U-shaped screw until the axis A1 of the outer surface of the pipeline **100** contacts the sensing surface S of the sensing element **10**, and the clamp **21** does not move along the axial direction of the pipeline **100** or rotate along the circumference of the pipeline **100**, it means that the clamp **21** has been correctly fixed on the outer surface of the pipeline **100**; that is, the center line of the bottom surface of the carrier **211** is aligned with the outer surface of the pipeline **100** axis A1 and the sensing surface S of the sensing element **10** is set on the installation datum plane P. Compared with the structure in which the carrier and the clamp are separated in the prior art, the carrier **211** of the present embodiment is a part of the clamp **21**, which reduces the screw-residual structure that will cause deformation of the carrier or clamp, and greatly reduces the possibility of installation failure and device failure.

In the process of fixing the clamp **21** and the pipeline **100**, since the slider **221** of the aligning member **22** is firstly placed on the outer surface of the pipeline **100**, the smaller the outer diameter of the pipeline is, the closer the outer surface of the pipeline **100** in contact with the A shape center of the slider **221** is, and the distance that the sensing surface S of the sensing element **10** moves to the outer surface of the pipeline **100** for fixing the clamp **21** is shorter. On the other hand, the larger the outer diameter of the pipeline **100** is, the closer the outer surface of the pipeline **100** in contact with the A shape opening of the slider **221** is, and the distance that the sensing surface S of the sensing element **10** moves to the outer surface of the pipeline **100** for fixing the clamp **21** will be longer. Therefore, the distance (D1-D2) that the aligning member **22** of the embodiment moves along the normal direction of the installation datum plane P during the installation of the clamp **21** is proportional to a size of the outer diameter of the pipeline **100**.

FIG. 3A is a side view of the fixing structure of the clamp-on type measuring device of a third embodiment of the present invention; FIG. 3B is a top cross-sectional view of the other aligning members shown in FIG. 3A; FIG. 3C is radial cross-section view of the fixing structure of the clamp-on type measuring device shown in FIG. 3A before fixed on the pipeline; FIG. 3D is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 3A after fixed on the pipeline. As shown in FIGS. 3A, 3B, 3C, and 3D, the fixing structure **3** of the clamp-on type pipeline measuring device includes: a clamp **31** and an aligning member **32**, the clamp **31** is detachably fixed on the outer surface of the pipeline **100** along the axis A1 of the outer surface of the pipeline **100**, and the aligning member **32** is disposed in the clamp **31**, wherein the plane tangent to the axis A1 with the outer surface of the pipeline **100** is defined as the installation datum plane P. After the clamp **31** is fixed on the outer surface of the pipeline **100**, the aligning member **32** contacts at least two line segments L5, L6 on the outer surface of the pipeline **100**, and the distances H5, H6 between the line segments L5, L6 and the installation datum plane P are the same.

In this embodiment, the clamp **31** includes a carrier **311** and a buckle **312**. The carrier **311** can be made of plastic or alloy molding. The carrier **311** can be made of plastic or alloy molding, is roughly rectangular, and can be equipped

with a sensing element **10** and a processing circuit (not shown), a power supply (not shown), a display screen **20**, etc. The sensing elements **10** can be ultrasonic probes, temperature sensors, vibration sensors, etc. The number of sensing elements **10** can be one or plural. The plurality of sensing elements **10** can be arranged in the carrier **311** along the long side direction of the carrier **311**, and the two sides of the long side of the carrier **311** have a buckle seat **3111** and an aligning groove **3112**. The buckle **312** include a polygonal seat **3121**, a spiral tube **3122**, a screw **3123**, and an elastic member **3124**, and can surround the outer surface of the pipeline **100** and connect to the buckle seat **3111** of the carrier **311**. The polygonal seat **3121** is roughly M-shaped, the two arms of the M-shape have channels, and one side of the spiral tube **3122** is embedded in the channel of the polygonal seat **3121**, and the inner walls of the two sides of the spiral tube **3122** have threads that can be engaged with the screws **3123**. The screw **3123** can be, but not limited to, contour screw. The elastic member **3124**, such as but not limited to spring, leaf spring, and rubber, is set between the polygonal seat **3121** and the screw **3123**. Two aligning members **32** are arranged on both sides of the long side of the carrier **311**, and the aligning member **32** includes a slider **321**, an elastic member **322** and a stop block **323**, and the slider **321** is roughly inverted U-shaped, the center of the inverted U-shape is aligned with the center point of the short side of the carrier **311**, and an accommodating groove **3211** is formed in the two arms of the U-shape. The two ends of the elastic member **322** are respectively inserted into the aligning groove **3112** of the carrier **311** and the accommodating groove **3211** of slider **321**. When the slider **321** moves away from the carrier **311**, the stop block **323** can prevent the aligning member **32** from moving out of the carrier **311**. As shown in FIGS. 3C and 3D, when the clamp **31** is not fixed on the pipeline, the slider **321** protrudes from the bottom surface of the carrier **311** by a length D3; after the clamp **31** is fixed on the outer surface of the pipeline **100**, the slider **311** protrudes from the bottom surface of the carrier **311** by a length D4, and the distance (D3-D4) that the slider **321** moves toward or away from the carrier **311** along the normal direction of the installation datum plane P ranges from 0.1 mm to 10 cm.

The process of installing the clamp **31** on the outer surface of the pipeline **100** includes: passing the screw **3123** through the buckle seat **3111** of the carrier **311**, placing the long side of the carrier **311** on the outer surface of the pipeline **100** along the axial direction of the pipeline **100** to make the slider **321** of the aligning member **32** abut against the four line segments on the opposite sides of the axis A1 of the outer surface of the pipeline **100**; embedding the spiral tube **3122** into the channels of the two arms of the polygonal seat **3121**, and using the polygonal seat **3121** and the spiral tube **3122** to surround the outer surface of the pipeline **100**; screwing the screws **3123** respectively into the two sides of the spiral tube **3122** to make the elastic member **3124** push the polygonal seat **3121** and the carrier **311** to move toward each other and to clamp the outer surface of the two opposite sides of the pipeline **100**, and concurrently to drive the outer surface of the pipeline **100** to push the aligning member **32** to gradually move toward the carrier **311** along the normal line vertical to the installation datum plane P (as Z-axis direction shown in FIG. 3C). Because the slider **321** of the aligning member **32** abuts against the two side segments L5, L6 of the axis A1 of the outer surface of the pipeline **100** and keeps the same distance from the installation datum plane P during the moving of the aligning member **32**, the center line of the bottom surface of the carrier **311** can be aligned with

the axis A1 of the outer surface of the pipeline **100** without deviation. When the screw **3123** is screwed until the axis A1 of the outer surface of the pipeline **100** contacts the sensing surface S of the sensing element **10**, and the carrier **311** and the polygonal seat **3121** will not move along the axial direction of the pipeline **100** or rotate along the circumference of the pipeline **100**, it means that the clamp **31** has been correctly fixed on the outer surface of the pipeline **100**; that is, the centerline of the bottom surface of the carrier **311** is aligned with the axis A1 of the outer surface of the pipeline **100**, and the sensing surface S of the sensing element **10** is set on the installation datum plane P.

In the process of fixing the clamp **31** and the pipeline **100**, since the slider **321** of the aligning member **32** is first placed on the outer surface of the pipeline **100**, the smaller the outer diameter of the pipeline is, the shorter the distance between the outer surface of the pipeline **100** and the sensing surface S of the sensing element **10** is, and the shorter the distance that the slider **321** moves to the fixing position will be. On the other hand, the larger the outer diameter of the pipeline **100** is, the longer the distance between the outer surface of the pipeline **100** and the sensing surface S of the sensing element **10** is, and the longer the distance that the slider **321** moves to the fixing position will be. Therefore, the distance (D1-D2) that the aligning member **22** of the embodiment moves along the normal direction of the installation datum plane P during the installation of the clamp **21** is proportional to a size of the outer diameter of the pipeline **100**. It is worth noting that the inverted U-shaped design of the slider **321** in this embodiment can not only ensure the direction and sensitivity of the moving of the alignment member **32** along the normal line of the installation datum plane P during the process of fixing the clamp **31**, but also effectively prevent the clamp **31** from rotating along the pipeline **100** during fixing.

FIG. 4A is a side view of the fixing structure of the clamp-on type measuring device of a fourth embodiment of the present invention; FIG. 4B is a bottom view of the other aligning members shown in FIG. 4A; FIG. 4C is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 4A. As shown in FIGS. 4A, 4B and 4C, the fixing structure **4** of the clamp-on type pipeline measuring device includes: a clamp **41** and an aligning member **42**, the clamp **41** is detachably fixed on the outer surface of the pipeline **100** along the axis A1 of the outer surface of the pipeline **100**, and the aligning member **42** is disposed in the clamp **41**, wherein the plane tangent to the axis A1 with the outer surface of the pipeline **100** is defined as the installation datum plane P. After the clamp **41** is fixed on the outer surface of the pipeline **100**, the aligning member **42** contacts at least two line segments L7, L8 on the outer surface of the pipeline **100**, and the distances H7, H8 between the line segments L7, L8 and the installation datum plane P are the same.

In this embodiment, the clamp **41** includes a carrier **411** and a buckle **412**. The carrier **411** can be made of plastic or alloy molding. The carrier **411** is roughly rectangular, and can be equipped with a sensing element **10**, a processing circuit (not shown), a power supply (not shown), a display screen **20**, etc. The sensing elements **10** can be ultrasonic probes, temperature sensors, vibration sensors, etc. The number of sensing elements **10** can be one or plural. The plurality of sensing elements **10** can be arranged in the carrier **411** along the long side direction of the carrier **411**, and the two sides of the long side of the carrier **411** have a buckle seat **4111**.

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The buckle **412** includes a base seat **4121**, a spiral tube **4122**, and a screw **4123**, and can surround the outer surface of the pipeline **100** and connect to the buckle seat **4111** of the carrier **411**. The base seat **4121** is an open shell, and a screw hole **41211** is formed in the center of the base seat **4121**. The aligning member **42** is arranged on the base seat **4121** of the buckle **412**, and includes a slider **421** and an adjusting member **422**. The slider **421** is roughly M-shaped, and the center of the M shape is aligned with the center of the base seat **4121**. The adjusting member **422** includes a knob **4221** and a threaded segment **4222**. The knob **4221** is set on a side of the base seat **4121** opposite to the slider **421**, the threaded segment **4222** passes through the screw hole **41211** of the base seat **4121** and connects to the M-shaped center of the slider **412**. The two arms of the M-shape of the slider **421** have sliding groove **4211** for the spiral tubes **4122** and the screw **4123** pass therethrough, and the slider **421** can move toward or away from the carrier **411** along the spiral tube **4122** for a distance of 0.1 mm to 10 cm according to the outer diameter of the pipeline **100** required.

The process of installing the clamp **41** on the outer surface of the pipeline **100** includes: placing the carrier **411** and the buckle **412** on opposite sides of the outer surface of the pipeline **100**, aligning the axis A2 of the outer surface of the pipeline **100** with the center of the base seat **4121**, placing the pipeline **100** on the slider **421** of the aligning member **42**, wherein the axes A1 and A2 of the outer surface of the pipeline **100** are located on the same semicircular section of the pipeline **100**; screwing the screws **4123** respectively to pass through the buckle **4111** and the through hole of the base seat **4121** then into the two sides of the spiral tube **4122** to fix the carrier **411** and the buckle **412**; rotating the adjusting member **422** to make the slider **421** gradually move toward the carrier **411** along the normal direction of the spiral tube **4122** (perpendicular to the installation datum plane P, as the Z-axis direction shown in FIG. 4C) and to push the axis A1 of the outer surface of the pipeline **100** against the sensing surface S of the sensing element **10**. When the clamp **41** does not move axially along the pipeline **100** or rotate along the circumference of the pipeline **100**, it means that the clamp **41** has been correctly fixed on the outer surface of the pipeline **100**; that is, the centerline of the bottom surface of the carrier **411** is aligned with the axis A1 of the outer surface of the pipeline **100** and the sensing surface S of the sensing element **10** is set on the installation datum plane P.

In the process of fixing the clamp **41** and the pipeline **100**, since the pipeline **100** is first placed on the slider **421** of the aligning member **42**, the smaller the outer diameter of the pipeline is, the longer the distance between the outer surface of the pipeline **100** and the sensing surface S of the sensing element **10** is, and the longer the distance that the slider **421** pushes the pipeline **100** to the fixing position will be. On the other hand, the larger the outer diameter of the pipeline **100** is, the shorter the distance between the outer surface of the pipeline **100** and the sensing surface S of the sensing element **10** is, and the shorter the distance that the slider **421** pushes the pipeline **100** to the fixing position will be.

Using the length D5 between starting point of the slider **421** movement and the fixed position of fixing the pipeline **100** as the distance that the aligning member **42** moves along the normal direction of the installation datum plane P during the installation of the clamp **41**, the distance (D5) that the aligning member **42** moves along the normal direction of the installation datum plane P during the installation of the clamp **41** is inversely proportional to a size of the outer diameter of the pipeline **100**. It is worth noting that when the

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carrier **411** and the buckle **412** are fixed by the spiral tube **4122** and the screw **4123**, the carrier **411** and the buckle **412** will not squeeze the pipeline **100**. By using the slider **421** made of a material in mechanical strength weaker than that of the pipeline **100** and the carrier **411**, to fix the clamp **41** and the pipeline **100** with the aligning member **42** abutting against the two sides of the outer surface of the pipeline **100**, even when the applied force is too large, the buckle **412** will only be enlarged to ensure that the carrier **411** and the buckle **412** will not squeeze the pipeline **100**, and the adjusting member **422** exerts a force on the center point of the slider **421**, so that uneven application of force does not occur, and shifting, warping or deforming of the clamp **41** and the pipeline **100** can be prevented.

FIG. 5A is a side view of the fixing structure of the clamp-on type measuring device of a fifth embodiment of the present invention; FIG. 5B is a top view of the other aligning members of FIG. 5A; FIG. 5C is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 5A before fixed on the pipeline; FIG. 5D is radial cross-sectional view of the fixing structure of the clamp-on type measuring device shown in FIG. 5A after fixed on the pipeline; and FIG. 5E is a radial schematic view of other aligning members of the fifth embodiment of the present invention.

As shown in FIGS. 5A, 5B, 5C and 5D, the fixing structure **5** of the clamp-on type pipeline measuring device includes: a clamp **51**, a first aligning member **52**, a second aligning member **53**, an outer diameter measuring unit **54**, and a limiter **55**. The clamp **51** is detachably fixed on the outer surface of the pipeline **100** along the axis A1 of the outer surface of the pipeline **100**, wherein the tangent plane passing through the axis A1 is defined as the installation datum plane P. After the clamp **51** is fixed on the outer surface of the pipeline **100**, the first aligning member **52** contacts at least two line segments L7, L8 of the outer surface of the pipeline **100**, and the second aligning member **53** contacts at least two line segments L5, L6 of the outer surface of the pipeline **100**. The distances H7, H8 between the line segments L7, L8 and the installation datum plane P are the same, and the distances H5, H6 between the line segments L5, L6 and the installation datum plane P are the same.

In this embodiment, the clamp **51** includes a carrier **511** and a buckle **512**. The materials and functions of the carrier **511** are as the carrier **411** of the fourth embodiment, and the carrier **511** can be equipped with a sensing element **10**, a processing circuit (not shown), a power supply (not shown), a display screen, etc. The two long sides of the carrier **511** have buckle seats **5111** and aligning grooves **5112**, one side of the carrier **511** has distance measuring grooves **5113**, and the center of the buckle seat **5111** is formed with a through hole. The buckle **512** includes a base seat **5121**, a spiral tube **5122**, and a screw **5123**, and can surround the outer surface of the pipeline **100** and connect to the buckle seat **5111** of the carrier **511**. The base seat **5121** is an open shell, and a screw hole **51211** is formed in the center of the base seat **5121**. The first aligning member **52** is disposed in the base seat **5121** of the buckle **512**, includes a first slider **521** and an adjusting member **522**. The first slider **521** is roughly M-shaped, and the center of the M-shape is aligned with the center of the base seat **5121**. The adjusting member **522** includes a knob **5221** and a threaded segment **5222**. The knob **5221** is arranged on the side of the base seat **5121** opposite to the first slider **521**. The threaded segment **5222** passes through the screw hole **51211** in the center of the base seat **5121** and connects to the center of the M-shape of the first slider **521**.

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The two arms of the M-shape have sliding grooves **5211** and guide grooves **5212**; the first slider **521** can move a distance of 0.1 mm to 10 cm along the spiral tube **5122** toward or away from the carrier **511** according to the outer diameter of the pipeline **100** required. The second aligning member **53** is arranged on both long sides of the carrier **511**, includes the second slider **531**, the elastic member **532**, and the stop block **533**. The second slider **531** is roughly in the shape of an inverted U-shape, and the center of the inverted U-shape is aligned with the center point of the short side of the carrier **511**. An accommodating groove **5311** is formed in the two arms of the U shape, and the two ends of the elastic member **532** (such as but not limited to spring, leaf spring, and rubber) are respectively embedded in the aligning groove **5112** and the accommodating groove **5311**. The second sliding block **531** can also move toward or away from the carrier **511** along the normal direction of the installation datum plane P for a distance of 0.1 mm to 10 cm according to the outer diameter of the pipeline **100** required, and the stop block **533** can prevent the second aligning member **53** from moving out of the carrier **511**. The outer diameter measuring unit **54** includes a distance measuring shaft **541** and a displacement sensor **542**. One end of the distance measuring shaft **541** is fixed on the second slider **531**, and the other end of the distance measuring shaft **541** is embedded in a distance measuring groove **5113** of the carrier **511**. The distance measuring shaft **541** moves with the second slider **531** in the distance measuring groove **5113**. The displacement sensor **542** is arranged in the carrier **511** adjacent to the distance measuring groove **5113**, and can generate a displacement data signal of the distance measuring shaft **541**. The limiter **55** is arranged between the buckle seat **511** and the first slider **521**, and has a protrusion **551**. The protrusion **551** is embedded in the guide groove **5212** of the first slider **521**, and can limit the movement distance of the first aligning member **52** to the carrier **511** along the normal direction of the installation datum plane P, so that the four screws **5123** can be evenly tightened without overlocking.

The process of installing the clamp **51** on the outer surface of the pipeline **100** includes: inserting the spiral tube **5122** into the limiter **55**, screwing the screw **5123** through the base seat **5121** and the first slider **521**, then into one side of the spiral tube **5122** to engage with the base seat **5121**, the first aligning member **52**, and the limiter **55**; placing the carrier **511** and the buckle **512** on the opposite sides of the outer surface of the pipeline **100**, so that the pipeline **100** is placed on the first slider **521** of the first aligning member **52**; screwing the screw **5123** through the buckle seat **5111**, then into the spiral tube **5122** to connect the carrier **511** and the buckle **512**; rotating the adjusting member **522** to make the first slider **521** move along the spiral tube **5122** and a protruding block **551** (the normal direction of the installation datum plane P, as the Z-axis direction shown in FIG. 5C), to push the outer surface of the pipeline **100** in contact with the second slider **531**, and to make the first slider **521** and the second slider **531** clamp at least four line segments on the outer surface of the pipeline **100**; continuing to rotate the adjusting member **522** so that the first slider **521** and the second slider **531** clamp the pipeline **100** and move along the normal direction of the installation datum plane P until the axis A1 of the outer surface of the pipeline **100** in contact with the sensing surface S of the sensing element **10**; when the clamp **51** does not move along the axial direction of the pipeline **100** or rotate along the circumference of the pipeline **100**, it means that the clamp **51** has been correctly fixed on the outer surface of the pipeline **100**; that is, the center

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line of the bottom surface of the carrier **511** is aligned with the axis A1 of the outer surface of the pipeline **100** and the sensing surface S of the sensing element **10** is set on the installation datum plane P.

The means for moving the first sliding block **521** is not limited to the above-mentioned adjusting member in the form of a bolt. FIG. 5E is a radial schematic view of other aligning members according to the fifth embodiment of the present invention. As shown in FIG. 5E, the first aligning member **52'** is disposed in the base seat **5121'** of the buckle **512**, the first aligning member **52'** includes a first slider **521** and an elastic member **523**, and the center of the base seat **5121'** has an accommodating groove **51211'**, the elastic member **523** (such as but not limited to spring, leaf spring, and rubber) is disposed in the accommodating groove **51211'** of the base seat **5121'** to connect the first slider **521**. The elastic member **523** can provide the elastic force for the first slider **521** to push the pipeline **100**. Since the elastic force of the elastic member **523** pushes the first slider **521** out of the base seat **5121'** for a certain length, the maximum vertical distance between the first aligning member **52'** and the second aligning member **53** is not greater than the outer diameter of the pipeline **100**. The carrier **511** and the buckle **512** are placed on the two sides of the pipeline **100** opposite to each other to make the first slider **521** and the second slider **531** contact the opposite sides of the outer surface of the pipeline. When the screws **5123** are screwed into the two sides of the spiral tube **5122** to fix the carrier **511** and the buckle **512**, the pipeline **100** presses the first slider **521** to move towards the base seat **5121'** and the second slider **531** to move towards the carrier **511** until the axis A1 of the outer surface of the pipeline **100** in contact with the sensing surface S of the sensing element **10**. The elastic forces generated by the extrusion of the elastic members **523**, **533** make the first slider **521** and the second slider **531** tightly clamp the opposite sides of the outer surface of the pipeline **100**, and fix the clamp **51** on the outer surface of the pipeline **100**.

In the process of fixing the clamp **51** and the pipeline **100**, since the pipeline **100** is first placed on the first slider **521** of the first aligning member **52**, the smaller the outer diameter of the pipeline **100** is, the longer the distance that the adjusting member **522** moves the first slider **521** to make the outer surface of the pipeline **100** contact the second slider **531** will be, or the longer the distance that the screw **5123** makes the first slider **521** move toward the base seat **5121'** will be, while the shorter the distance that the second slider **531** moves until the outer surface of the pipeline **100** abuts against the sensing surface S of the sensing element **10** will be when the outer surface of the pipeline **100** contacts the second slider **531**. On the other hand, the larger the outer diameter of the pipeline **100** is, the shorter the distance that the adjusting member **522** moves the first slider **521** to make the outer surface of the pipeline **100** contact the second slider **531** will be, or the longer the distance that the screw **5123** makes the first slider **521** move toward the base seat **5121'** will be, while the longer the distance that the second slider **531** moves until the outer surface of the pipeline **100** abuts against the sensing surface S of the sensing element **10** will be when the outer surface of the pipeline **100** contacts the second slider **531**. Using the lengths D6, D7 between the starting point of the first slider **521** moving away from the base seat **5121** or toward the base seat **5121'** to the fixed position of the fixed pipeline **100** as the movement distances of the first alignment member **52**, **52'** along the normal direction of the installation datum plane P during the installation process of clamp **51**, the movement distances of the

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first alignment member **52**, **52'** along the normal direction of the installation datum plane P during the installation process of clamp **51** is inversely proportional to a size of the outer diameter of the pipeline **100**. Using the length difference that the second slider **531** protrudes beyond the bottom surface of the carrier **511** before touching the outer surface of the pipeline **100** and after fixed as the movement distance of the second alignment member **53** along the normal direction of the installation datum plane P, the movement distance of the second alignment member **53** along the normal direction of the installation datum plane P is proportional to a size of the outer diameter of the pipeline **100** during the installation process of clamp **51**.

By using the linear movement distance of the aligning member is proportional or inversely proportional to a size of the outer diameter of the pipeline, the distance measuring shaft **541** of the outer diameter measuring unit **54** is linked with the second aligning member **531** and is provided with optical or magnetic elements, the displacement sensor **542** of the outer diameter measuring unit **54** senses the moving distance of the distance measuring shaft **541**, so as to calculate the outer diameter of the pipeline, which can be used as basic parameter for the clamp-on type pipeline measuring device to calculate the physical quantities (such as, flow velocity, flow rate, temperature) of the fluid in the pipeline.

It is worth noting that when fixing the carrier **511** and the buckle **512** with the spiral tube **5122** and the screw **5123** and fixing the clamp **51** and the pipeline **100** with the first aligning member **52**, the limiter **55** disposed between the buckle seat **5111** of the carrier **511** and the first slider **521** of the first aligning member **52** can prevent the excessive application of force by individual screws **5123** and the first aligning member **52**, and the application of uneven force to different screws **5123** and the first aligning member **52**, so as to ensure that the clamp **51** and the pipeline **100** will not be deflected, warped or deformed.

In the process of fixing the clamp and the pipeline, since the aligning member can keep the clamp and the pipeline relatively moving along the normal direction of the installation datum plane until the sensing surface of the sensing element is fixed on the axis of the outer surface of the pipeline, the present invention is not limited to screw locking methods, such as screws, nuts, and spiral tubes.

FIG. 6A is a side view of the fixing structure of the clamp-on type measuring device of a sixth embodiment of the present invention; FIG. 6B is an axial view of a fixing structure of the clamp-on type measuring device shown in FIG. 6A. As shown in FIGS. 6A and 6B, the fixing structure **6** of the clamp-on type pipeline measuring device includes a clamp **61** and an aligning member **62**. The clamp **61** includes a carrier **611** and a buckle **612**. The material, structure, and function of the carrier **611** and the aligning member **62** are the same as those described in the first to third embodiments, the aligning member **62** is disposed in the carrier **611**. During the processes of detaching and fixing the clamp **61**, the aligning member **62** moves along the normal direction of the installation datum plane P. After the clamp **61** is fixed on the outer surface of the pipeline **100**, the aligning member **62** contacts at least two line segments L9 on the outer surface of the pipeline **100**, and the distances H9 between the two line segments L9 and the installation datum plane P are the same. In this embodiment, the buckle **612** is a belt, which surrounds the outer surface of the pipeline **100** and connects to the buckle seat **6111** of the carrier **611**, so that the clamp **61** is fixed on the pipeline **100**.

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FIG. 7A is a side view of the fixing structure of the clamp-on type measuring device of a seventh embodiment of the present invention; FIG. 7B is a radial view of a fixing structure of the clamp-on type measuring device shown in FIG. 7A. As shown in FIGS. 7A and 7B, the fixing structure **7** of the clamp-on type pipeline measuring device includes a clamp **71** and an aligning member **72**, and the clamp **71** includes a carrier **711** and a buckle **712**. The materials, structure and functions of the carrier **711** and an aligning member **72** are the same as described in the first to third embodiments; the aligning member **72** is disposed in the carrier **711**. During the processes of disassembling and fixing the clamp **71**, the aligning member **72** moves along the normal direction of the installation datum plane P. After the clamp **71** is fixed on the outer surface of the pipeline **100**, the aligning member **72** contacts at least two line segments L10, L11 on the outer surface of the pipeline **100**, and the distances H10, H11 between the line segments L10, L11 and the installation datum plane P are the same. In this embodiment, the buckle **712** is a pipe clamp, and the buckle seat **7111** connected to the carrier **711** can clamp the outer surface of the pipeline **100** so that the clamp **71** is fixed on the outer surface of the pipeline **100**.

It is worth noting that the above-mentioned embodiments show and illustrate that the sensing element is arranged in the carrier, and the carrier is set on one side of the outer surface of the pipeline. However, other embodiments wherein the sensing element is arranged in the clamp, or the two opposite sides of the outer surface of the pipeline are provided with the carriers, a plurality of sensing elements are arranged in the two carriers, and the buckles surround the outer surface of the pipeline and connects the two carriers, are also within the scope of the technical content of the present invention.

The present invention provides a clamp-on type pipeline measuring device, including a fixing structure and a sensing element as described in the above-mentioned embodiments, the fixing structure includes a clamp and an aligning member, wherein the clamp is detachably fixed on the outer surface of the pipeline, the aligning member is disposed in the clamp; the sensing element is arranged in the clamp and has a sensing surface. When the outer surface of the pipeline contacts the sensing surface, the clamp is fixed on the outer surface of the pipeline, the aligning member contacts at least two line segments on the outer surface of the pipeline, and the distances between the two line segments and the sensing surface are the same. As described in the above-mentioned first to seventh embodiments, the clamp includes the carrier and the buckle, the aligning member is disposed in the carrier or the buckle, and the sensing element is at least one of ultrasonic probe, temperature sensor, and vibration sensor.

In summary, in the fixing structure of the clamp-on type pipeline measuring device of the present invention, the aligning member can be applied to pipelines with different pipe diameters, especially pipes with the same nominal diameter of different pipeline standards. In the process of fixing the clamp and the pipeline, the aligning member moves along the normal direction of the tangent plane which passes through the axis of the outer surface of the pipeline, so that the center line of the clamp can be aligned with the axis of the outer surface of the pipeline. After the clamp is fixed on the outer surface of the pipeline, the aligning member contacts at least two line segments equidistant from the outer surface of the pipeline and the installation datum plane, which can keep the clamp in the correct installation position, so that the sensing surface of the sensing element

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is set on the outer surface of the pipeline on the tangent plane of the axis. The clamp-on type pipeline measuring device of the present invention reduces the use of screws to adjust the residual between the clamp and the pipeline, and effectively prevents the clamp, pipeline, and sensing elements from shifting, warping, deforming and being damaged due to improper installation, and thus, improves the convenience of installation and ensure the correctness of the measurement of the clamp-on type pipeline measuring device.

Although the present invention has been described with reference to the preferred embodiments thereof, it is apparent to those skilled in the art that a variety of modifications and changes may be made without departing from the scope of the present invention which is intended to be defined by the appended claims.

What is claimed is:

1. A fixing structure of a clamp-on type pipeline measuring device, comprising:

a clamp, detachably fixed on an outside surface of a pipeline along an axis of the outer surface of the pipeline; and

an aligning member, disposed in the clamp, wherein a plane tangent to the outer surface of the pipeline on the axis is defined as an installation datum plane; during detaching and fixing the clamp, the aligning member moves along a normal direction of the installation datum plane; after the clamp is fixed to the outer surface of the pipeline, the aligning member contacts at least two line segments on the outer surface of the pipeline, and distances between the two line segments and the installation datum plane are the same.

2. The fixing structure of the clamp-on type pipeline measuring device according to claim 1, wherein the clamp comprises a carrier and a buckle, the carrier is set on the installation datum plane, and the buckle surrounds the outer surface of the pipeline and connects to the carrier.

3. The fixing structure of the clamp-on type pipeline measuring device according to claim 2, wherein the aligning member is arranged in the carrier, and a distance that the aligning member moves along the normal direction of the installation datum plane is proportional to a size of an outer diameter of the pipeline.

4. The fixing structure of the clamp-on type pipeline measuring device according to claim 2, wherein the aligning member is arranged in the buckle, and a distance that the aligning member moves along the normal direction of the

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installation datum plane is inversely proportional to a size of an outer diameter of the pipeline.

5. The fixing structure of the clamp-on type pipeline measuring device according to claim 4, further comprising an outer diameter measuring unit, arranged on the carrier; wherein the outer diameter measuring unit moves along the normal direction of the installation datum plane during the detaching and fixing the clamp, and a distance that the outer diameter measuring unit moves along the normal direction of the installation datum plane is proportional to the size of the outer diameter of the pipeline.

6. The fixing structure of the clamp-on type pipeline measuring device according to claim 1, wherein the aligning member comprises a slider, the slider has two protruding arms, and a surface between the two arms is in the shape of a single angle, a polygon, a plane, or an arc.

7. The fixing structure of the clamp-on type pipeline measuring device according to claim 6, wherein the aligning member further comprises an elastic member, and two ends of the elastic member are respectively embedded in the clamp and the slider, so as to generate an elastic force for moving the aligning member along the normal direction of the installation datum plane.

8. A clamp-on type pipeline measuring device, comprising:

the fixing structure according to claim 1 comprising: the clamp and the aligning member, wherein the clamp is detachably fixed on an outer surface of a pipeline, the aligning member is disposed in the clamp; and

a sensing element, arranged in the clamp and having a sensing surface; wherein when the outer surface of the pipeline contacts the sensing surface, the clamp is fixed on the outer surface of the pipeline, the aligning member contacts at least two line segments of the outer surface of the pipeline, and distances between the two line segments and the sensing surface are the same.

9. The clamp-on type pipeline measuring device according to claim 8, wherein the clamp comprises a carrier and a buckle, and the aligning member is arranged in the carrier and/or the buckle.

10. The clamp-on type pipeline measuring device according to claim 8, wherein the sensing element is at least one of an ultrasonic probe, a temperature sensor, and a vibration sensor.

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